Less Architecture, More Recovery: A Training-Free Inference Framework and a Pretraining-Regime Study for SAR Ship Instance Segmentation

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***Abstract*—Deep learning methods have substantially advanced ship instance segmentation in synthetic aperture radar (SAR) imagery. However, the residual error concentrates in the near-shore regime, where small, low-contrast hulls are embedded in bright dock clutter and specialized methods lose 25 to 30 mask average precision (AP) points relative to their own offshore performance. This article advances that regime without any new architecture. On HRSID, the proposed pipeline reaches 44.5 inshore mask AP and 48.3 inshore AP_{75}, above the strongest published 500-epoch specialized architecture on both metrics after 55 epochs of training, while its training-free rung alone attains an inshore AP_{50} of 75.6 against 69.5 for that architecture. The margins come from treating the pretraining regime, not the architecture, as the largest lever: the same Mask2Former scored 17.9 inshore AP from scratch but 27.7 from Common Objects in Context (COCO) initialization, whereas seven architecture- and training-side modifications of that base were flat, negative, or absorbed. We therefore leave the model untouched and build a three-rung inference cascade. The first rung is training-free and composes three operators at disjoint pipeline stages, a same-basin weight soup in weight space, wide multi-scale flip test-time augmentation (TTA) in input space, and a consensus-gated mask intersection-over-union (IoU) vote fusion in output space, lifting 27.7/45.6/59.6 to 38.8/53.8/64.8 mask AP on the inshore, official, and offshore splits at zero additional training and roughly $8\times$ inference cost. Controlled ablations identify the fusion operator, not the view count, as the enabler of aggressive TTA: the identical views collapse to 4.1 inshore AP under matrix non-maximum suppression (NMS) and erode offshore AP to 55.9 under pixelwise majority voting. The remaining two rungs are self-correctors fitted on the base model's own training-split errors and need no new annotation. A 2.35M-parameter boundary-patch refiner, trained in about 19 min and confined to a ± 4-pixel band around predicted boundaries, raises inshore AP_{75} from 36.1 to 47.0; a 14-feature gradient-boosted consensus re-ranker, fitted in 19 s, then reorders detections without editing a single mask pixel and carries the three splits from 43.6/58.6/69.1 to 44.5/59.4/70.1 mask AP. Across six further carriers spanning five architecture families, the complete recipe lifts every carrier on every split. Experiments on HRSID and PSeg-SSDD demonstrate the effectiveness and robustness of the proposed framework.**

***Index Terms*—Instance segmentation, model soup, ship detection, synthetic aperture radar (SAR), test-time augmentation (TTA).**

I. Introduction

With its all-weather, day-and-night imaging capability, synthetic aperture radar (SAR) plays a key role in maritime surveillance, fisheries monitoring, and vessel traffic management. Ship instance segmentation extends detection from bounding boxes to pixel-accurate masks and thereby supplies the hull geometry that heading estimation, berthing analysis, and fine-grained recognition require. The release of pixel-annotated benchmarks such as HRSID [1] has made the task quantitatively comparable across methods, and deep segmentation networks now dominate the reported results.

Progress, however, splits between two regimes. Offshore, well-resolved ships on open water are handled adequately by generic instance segmentation models. Inshore, small hulls with low or even negative contrast are buried in bright, structured dock clutter under single-look speckle, and the specialized architectures published to date lose 25 to 30 mask average precision (AP) points relative to their own offshore performance, while classic detectors and from-scratch baselines lose more than 30. The open problem therefore lives almost entirely in the near-shore split.

This article shows that the near-shore split advances furthest without a new architecture. On HRSID, the proposed pipeline reaches 44.5 inshore mask AP and 48.3 inshore AP_{75}, above the strongest published 500-epoch specialized architecture (O2Former [2], 43.1 and 45.9) on both metrics after 55 epochs of training; its training-free tier alone attains an inshore AP_{50} of 75.6 against O2Former's 69.5; and the composed pipeline lifts its own base by +16.8 inshore AP.

The prevailing response in the literature has been architectural specialization. O2Former augments Mask2Former with an optimized query generator and an orientation-aware polar-coordinate embedding module [2], DiffSARShipInst couples a diffusion-based instance decoder with asymmetric-convolution mask refinement [3], a bounding-box-free formulation encodes each ship as an elliptical Gaussian heatmap under centroid- and orientation-constrained losses [4], LFG-Net routes low-level feature guidance through super-resolution denoising to counter near-shore scattering [5], and SARSAM transfers the Segment Anything Model to SAR by parameter-efficient tuning [6]. O2Former and the baselines re-tabulated in its benchmark share a common protocol, namely, 500-epoch training with an ImageNet backbone and a from-scratch segmentation head, and all of these works attribute their gains to new operators. None of them isolates the pretraining regime as a variable.

This article starts from a different measurement. Under the from-scratch-head protocol, plain Mask2Former [7] scores 17.9 inshore mask AP on HRSID [2]; the identical architecture

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initialized from Common Objects in Context (COCO) instance segmentation weights reaches 27.7 after only 55 epochs (three-seed mean). The pretraining regime alone is worth 9.8 points inshore, more than any single architectural component ablated in the specialized literature.

On this strong base we then ran a controlled seven-direction architecture- and training-side sweep, and no direction paid. Constraining the deformable-attention sampling offsets cost 1.0 inshore point, one-to-many auxiliary supervision was flat, four training-side families (capacity, feature enrichment, data augmentation, and supervision) were each subsumed by the COCO-initialized base, and a mask intersection-over-union (IoU) rescoring head retained only a +0.1 residual once the weight soup was applied. A zero-gated strip adapter, built from sequential $1 \times 7/7 \times 1$ depthwise convolutions in line with current strip-operator guidance [8]–[10] and initialized to preserve the COCO weights exactly in the spirit of LayerScale [11], cost 0.9 inshore point, and a Swin-S backbone swap [12] gained +0.9 at the base yet fell below the ResNet-50 configuration on all three splits once the full recovery stack was applied. This ledger echoes the preservation evidence from the adaptation literature, where adapters that protect strong pretrained representations outperform full fine-tuning [13] and frozen foundation features enrich detectors without being overwritten [14]. The residual inshore error does not behave as a capacity problem.

If the residual is instead dominated by variance (which basin the short schedule lands in, which view of the scene the model sees, and how overlapping predictions are merged), then the matched remedy is training-free. We recover accuracy with three operators at three disjoint pipeline stages. In weight space, a same-basin soup averages three independently seeded checkpoints elementwise [15] and adds +1.9 inshore at $1\times$ inference cost; reported weight-averaging gains concentrate in from-scratch regimes [16], [17], and fine-tuning from a strong checkpoint is exactly where that literature is thinnest. In input space, wide multi-scale flip test-time augmentation (TTA) over eight views places more pixels on pixel-starved near-shore hulls: the six-view core adds +6.9 inshore, robust across seeds, and widening the scale set to $\{640, 800, 960, 1088\}$ adds a further +2.0. In output space, a consensus-driven vote fusion clusters candidates by true mask IoU rather than box IoU and, gated by cross-view agreement, emits either the seed mask or a consensus-denoised mask per cluster. The first two operators stack near-additively (predicted 36.5 inshore, measured 36.1). The third is not an additive term but an enabler, because the same eight views collapse to 4.1 inshore under matrix non-maximum suppression (NMS) and erode offshore AP from 59.6 to 55.9 under unconditional pixelwise majority voting. Composed, the stack lifts 27.7/45.6/59.6 to 38.8/53.8/64.8 mask AP on the inshore, official, and offshore splits (+11.1/+8.2/+5.2) at zero additional training and roughly $8\times$ inference. The exchange is explicit. A training budget is spent instead as a deliberation budget, and the accuracy-compute dial that results stays in the operator's hands rather than being frozen into a set of weights.

We position the result as a cascade of three rungs and claim neither more nor less than the numbers support. The training-free rung posts the strongest detection axis, an inshore AP_{50} of 75.6 against O2Former's 69.5, although it does not lead on absolute inshore AP (38.8 against 43.1) or AP_{75} (36.1 against 45.9); the stack is a detection-and-recall recovery, not a boundary-precision method. What the two remaining rungs share is their supervision. A deployed model makes its mistakes on the training split as readily as on the test split, and those mistakes are free supervision: we train nothing but the corrector, and no new annotation is drawn.

The second rung supplies exactly the missing ingredient as a post-processor in the boundary-patch refinement (BPR) paradigm [18]. A 2.35M-parameter UNet, trained in about 19 min on train-split boundary patches, edits masks only within a ± 4-pixel band around predicted boundaries and never touches the main model or any detection score. With this refiner, inshore reaches 43.6 AP and 47.0 AP_{75} under the protocol caveats stated in Section IV. The third rung closes the ranking axis. Mask AP is a ranking metric, and a candidate whose mask is accurate but whose score sits below a worse neighbor forfeits precision at every recall level it should have dominated; the fusion stage already computes the quantity that predicts mask quality, namely the cross-view consensus of each cluster, yet passes the seed score through unchanged, because no hand-written function of that quantity helped all three splits at once. We therefore learn the function rather than write it down. A gradient-boosted regression tree over 14 scalar features, fitted in 19 s on the base model's own training-split errors, re-ranks detections without editing a single mask pixel and carries the pipeline to 44.5 inshore AP and 48.3 inshore AP_{75}, above the 500-epoch O2Former on both metrics at 55 epochs of training, with 59.4 official and 70.1 offshore, where that architecture still leads by 1.6. The training-free claim is accordingly scoped to the recovery stack, and both correctors are reported separately throughout. An equal-epoch control reinforces this framing: extending training to 200 epochs and applying the same recovery yields 37.9/52.5/63.5, below the short-schedule configuration. Fig. 1 summarizes the framework and the pipeline stage at which each operator acts.

The main contributions of this article are summarized as follows.

- A pretraining-regime study for SAR ship instance segmentation. COCO initialization alone moves plain Mask2Former from 17.9 to 27.7 inshore mask AP, and a controlled seven-direction ablation shows that every architecture- or training-side modification of the strong base was flat, negative, or absorbed, a pattern we term the preservation regularity.
- A training-free three-operator recovery framework, composed of a same-basin weight soup, wide multi-scale flip TTA, and consensus-vote fusion, that lifts inshore mask AP by +11.1 (+40% relative) at zero additional training and roughly $8\times$ inference cost. Transferred to PSeg-SSDD, the same stack exceeds six comparison methods that we trained ourselves under one protocol, drawn from the two-stage, single-stage prototype, dynamic-kernel, and SAR-specific module families, on all three splits.
- Evidence that the fusion operator, not the view count, decides whether aggressive TTA helps or harms. Under

identical views, matrix NMS collapses to 4.1 inshore and majority voting erodes offshore AP below the no-TTA base, whereas the consensus vote matches or exceeds the seed variant on all three splits.
- A negative-results ledger reported under the same controls as the positives, including the seven-direction sweep and a 200-epoch equal-budget null, which rules out capacity as the recoverable lever in the forms tested.
- A three-rung self-correcting inference cascade that adds two small trained correctors on top of the training-free stack, each fitted on the base model's own errors on the training split and neither requiring a new annotation. The first is a 2.35M-parameter boundary-patch refiner, trained in about 19 min and confined to a ± 4-pixel boundary band, which converts the stack's known boundary imprecision into accuracy (inshore AP_{75} from 36.1 to 47.0). The second is a 14-feature gradient-boosted consensus re-ranker, fitted in 19 s on statistics the fusion stage already produces, which edits detection scores only and leaves every mask pixel bit-identical, carrying the pipeline to 44.5 inshore AP and 48.3 inshore AP_{75}, above the strongest published 500-epoch architecture on both metrics. The refiner is audited along seven ground-truth-leakage vectors and the re-ranker under a separate adversarial protocol, and applying the recipe to six further carriers spanning five architecture families across two datasets leaves every carrier above its own baseline on all three splits.

The rest of this article is organized as follows. Section II reviews related work. Section III details the recovery framework and the two optional correctors. Section IV presents experiments and ablations on HRSID and the cross-dataset transfer to PSeg-SSDD. Section V discusses the preservation regularity and the limitations of the study, and Section VI concludes this article.

II. Related Work

A. SAR Ship Instance Segmentation

Published SAR ship instance segmentation methods are predominantly specialized architectures trained from scratch, that is, an ImageNet-pretrained backbone with randomly initialized detection and mask heads, under long schedules on HRSID and SSDD. LFG-Net exploits low-level feature guidance with super-resolution denoising to counter the scattering and noise characteristics of near-shore scenes [5], a bounding-box-free Gaussian-heatmap formulation with centroid- and orientation-constrained losses reports 42.5 inshore mask AP on HRSID [4], and DiffSARShipInst couples a diffusion-based instance decoder with aspect-ratio-matched asymmetric-convolution mask refinement and reports 42.6 inshore [3], although its published figures circulate through secondary tables and have, to our knowledge, not been independently reproduced. O2Former holds the strongest split-matched inshore result to date, 43.1 mask AP, by augmenting Mask2Former [7] with an optimized query generator and an orientation-aware polar-coordinate module under a 500-epoch from-scratch protocol in which its own Mask2Former baseline reaches only 17.9 inshore [2]. SARSAM instead adapts the Segment Anything Model by parameter-efficient tuning, but reports only AP at IoU 0.50 (90.12 on the full HRSID test set), which precludes full-range COCO-AP comparison [6]. No published method isolates the pretraining regime as a variable. We show that COCO initialization alone moves the same Mask2Former from 17.9 to 27.7 inshore mask AP, and we study training-free recovery on that strong base rather than a new architecture.

B. Parameter-Efficient Adaptation on Strong Bases

A growing adapter literature argues that strong pretrained representations should be preserved rather than overwritten. Mona shows that a multi-cognitive convolutional adapter surpasses full fine-tuning on COCO instance segmentation by +0.9 mask AP with a Swin-B Cascade Mask R-CNN, whereas the linear low-rank adapter LoRand falls 0.4 AP below full fine-tuning (44.7 versus 45.1), so both preservation and the adapter's inductive content matter [13]. Frozen-DETR likewise enriches a detector with tokens from a frozen vision foundation model and gains +2.9 AP on natural-image benchmarks [14]. Our findings sharpen this principle for SAR: on a COCO-pretrained Mask2Former, seven controlled architecture- and training-side arms, including a zero-initialized gated convolutional adapter that preserves the pretrained function exactly at the start of fine-tuning, were flat, negative, or insufficient (Section IV). Preservation alone does not guarantee a gain. The injected prior must also match the target regime (Section II-D).

C. Test-Time Augmentation and Weight Ensembling

Weight averaging is standard practice in detection: exponential moving average is a routine training configuration in modern detectors [19], with reported gains of up to +2.5 AP in from-scratch training but only +0.2 to +0.6 when fine-tuning strong checkpoints [16]; model soups average the weights of several models fine-tuned from a shared initialization [15]; and a recent empirical study of weight averaging concentrates its experiments on from-scratch regimes [17], leaving fine-tuning from strong checkpoints under-quantified. Multi-scale flip test-time augmentation (TTA) is commonly credited with +0.5 to +1.5 AP on COCO, whereas we observe +6.9 inshore mask AP from a six-view ensemble. On the output side, BPR crops patches along predicted instance contours and re-segments them with a small high-resolution network as a model-agnostic post-processing stage [18]; we adopt this paradigm as an optional 2.35M-parameter second tier and audit it for ground-truth leakage (Section IV). Re-ranking the outputs of a frozen detector belongs to the same inference-side family, and its nearest precedent in detection is DiffusionDet, which turns the number of proposal boxes and denoising steps into an evaluation-time dial that buys accuracy with computation on an already-trained model, 45.8 to 46.8 box AP on COCO [20]. That dial spends extra compute on generating candidates; our third tier spends it on adjudicating them, learning a mask-quality score from the consensus statistics that the multi-view stack produces anyway and re-ordering detections whose masks it never touches. What this literature leaves open is the fusion question. Our ablations show that the merge operator,

not the view count, decides whether aggressive TTA helps or collapses, and that a same-basin weight soup matches a three-seed prediction ensemble at one third of the inference cost.

D. Anisotropic Priors

Elongated targets have motivated strip-shaped operators. SegNeXt employs multiplicative strip attention [9], InceptionNeXt uses parallel strip kernels with a band-kernel sweet spot near $k = 11$ [10], and Strip R-CNN applies sequential $1 \times k/k \times 1$ strip convolutions to remote-sensing detection [8]. Learnable per-channel residual scaling with small initialization, introduced as LayerScale for deep transformers [11], informs the initialization of such adapters, and in SAR the orientation-aware module of O2Former encodes anisotropy directly [2]. Our controlled test found that the strip prior transfers negatively to the inshore split (-0.9 mask AP) and marginally positively offshore (+0.6): small near-shore HRSID ships are quasi point-like, so a plausible elongation prior fails under regime shift. We report this negative transfer as controlled evidence rather than discard it.

III. Method

This section presents the proposed framework. We first fix notation and the problem setup, then describe the base model and the pretraining regime on which every later number rests. The three training-free recovery operators follow in pipeline order: a same-basin weight soup in weight space (Sec. III-C), wide multi-scale flip test-time augmentation (TTA) in input space (Sec. III-D), and a consensus-gated mask-IoU vote fusion in output space (Sec. III-E). We close with two optional self-correctors, a boundary-patch refiner that may edit only mask pixels (Sec. III-F) and a learned consensus re-ranker that may edit only scores (Sec. III-G), and with an analysis of computational cost (Sec. III-H). Fig. 1 summarizes the pipeline. The figures of this section draw measured tensors rather than sketches, and three conventions hold across all of them. Every tensor is drawn to one geometric scale, 1 mm of printed edge per 40 image pixels and $+0.55$ mm of slab depth per channel doubling; the only exception is the $2048@25^2$ backbone slab of Fig. 1, drawn at a 1.2 mm legibility floor instead of its true 0.63 mm and marked $\approx$. At that scale the median HRSID inshore ship, 0.028% of an 800×800 chip or 13.4 px across, measures 0.33 mm, which is why the boundary crops of Fig. 5 are shown in explicit zooms. Feature maps are rendered as the L_2 norm over channels in *viridis* and are upsampled by nearest neighbor, so no spatial detail is invented, and the provenance of every drawn chip, tensor, and matrix is recorded in `figs/assets/manifest.json`.

A. Problem Setup and Notation

Given a single-look SAR amplitude chip $\mathbf{x} \in \mathbb{R}^{B\times 3\times 800\times 800}$, in which the grayscale channel is replicated to three channels and $B = 1$ at test time, ship instance segmentation asks for a set of scored binary masks

$$\mathcal{Y} = \big\{(\mathbf{m}_i, s_i)\big\}_{i=1}^{N_{\text{out}}}, \quad \mathbf{m}_i \in \{0,1\}^{800\times 800},\ s_i \in [0,1], \quad (1)$$

where $\mathbf{m}_i$ is the binary mask of the ith instance, s_i its confidence, and $N_{\text{out}} \le 100$ after score ranking under the COCO convention. Evaluation is COCO mask AP ($\times 100$ throughout) on the HRSID official test split and on its inshore and offshore partitions, following the protocol of [2]. Tensors are ordered $B \times C \times H \times W$; bold lower case denotes vectors or masks, bold upper case matrices, calligraphic letters sets, and θ the full parameter vector of the segmentation network Φ_θ. The method has two stages, and its inference side is a cascade of three rungs. Stage A is a standard, unmodified Mask2Former-R50 [7] fine-tuned from COCO initialization. Stage B, our contribution, updates no weight of Stage A. Its first rung is training-free and acts at three disjoint pipeline stages, weight space, input space, and output-fusion space. Its second and third rungs are two small correctors fitted on Stage A's own errors on the training split rather than on new annotations: a boundary-patch refiner that may write only mask pixels (Sec. III-F) and a consensus re-ranker that may write only scores (Sec. III-G). Every rung is optional, every rung is reported separately in each table, and the training-free claim of this article is scoped to the first.

B. Base Model and the Pretraining Regime

The base is used exactly as released. The ResNet-50 backbone produces a four-level pyramid $\{C_2, C_3, C_4, C_5\}$ at strides $\{4, 8, 16, 32\}$, with shapes $B \times 256 \times 200 \times 200$, $B\times 512\times 100\times 100$, $B\times 1024\times 50\times 50$, and $B\times 2048\times 25\times 25$ for an 800×800 input. A six-layer multi-scale deformable-attention pixel decoder (8 heads, 256 dimensions) fuses C_3–C_5 into a stride-4 mask feature $\mathbf{F} \in \mathbb{R}^{B\times 256\times 200\times 200}$ plus three flattened memory levels at strides $\{8, 16, 32\}$. $N = 100$ learnable queries $\mathbf{Q} \in \mathbb{R}^{B\times 100\times 256}$ traverse nine decoder layers, each applying masked cross-attention to one memory level in round-robin order, query self-attention, and a feed-forward block. A linear head maps each query to class logits in $\mathbb{R}^{B\times 100\times 2}$ (ship versus no-object); a mask head maps it to an embedding $\mathbf{e}_q \in \mathbb{R}^{256}$, and the mask of query q is read out as

$$\mathbf{m}_q = \mathbf{1}\Big[\sigma\big(\mathcal{U}_{4\times}(\mathbf{e}_q^\top \mathbf{F})\big) \ge 0.5\Big], \qquad q = 1, \ldots, 100, \quad (2)$$

where $\mathbf{e}_q^\top \mathbf{F} \in \mathbb{R}^{200\times 200}$ is the per-pixel dot product between the mask embedding and $\mathbf{F}$, $\mathcal{U}_{4\times}$ denotes bilinear upsampling to 800×800, σ is the sigmoid, and $\mathbf{1}[\cdot]$ is the indicator applied pixelwise.

Stage A fine-tunes this model from COCO instance-segmentation weights for 55 epochs (100,872 iterations at batch size 2 with 2-step gradient accumulation) at 800×800 resolution and initial learning rate 2.5×10^{-5} on a single RTX 3090 Ti. An auxiliary mask-IoU rescoring head present in the implementation is disabled ($\lambda = 0$); the weight soup subsumes its gain (Section IV). The initialization is not a detail. Under the 500-epoch from-scratch-head protocol of [2], the identical architecture reaches 17.9 inshore mask AP; with COCO initialization it reaches 27.7 (three-seed mean) after 55 epochs. This 9.8-point regime gap exceeds any single architectural increment reported in the specialized SAR instance-segmentation literature. We therefore impose a preservation

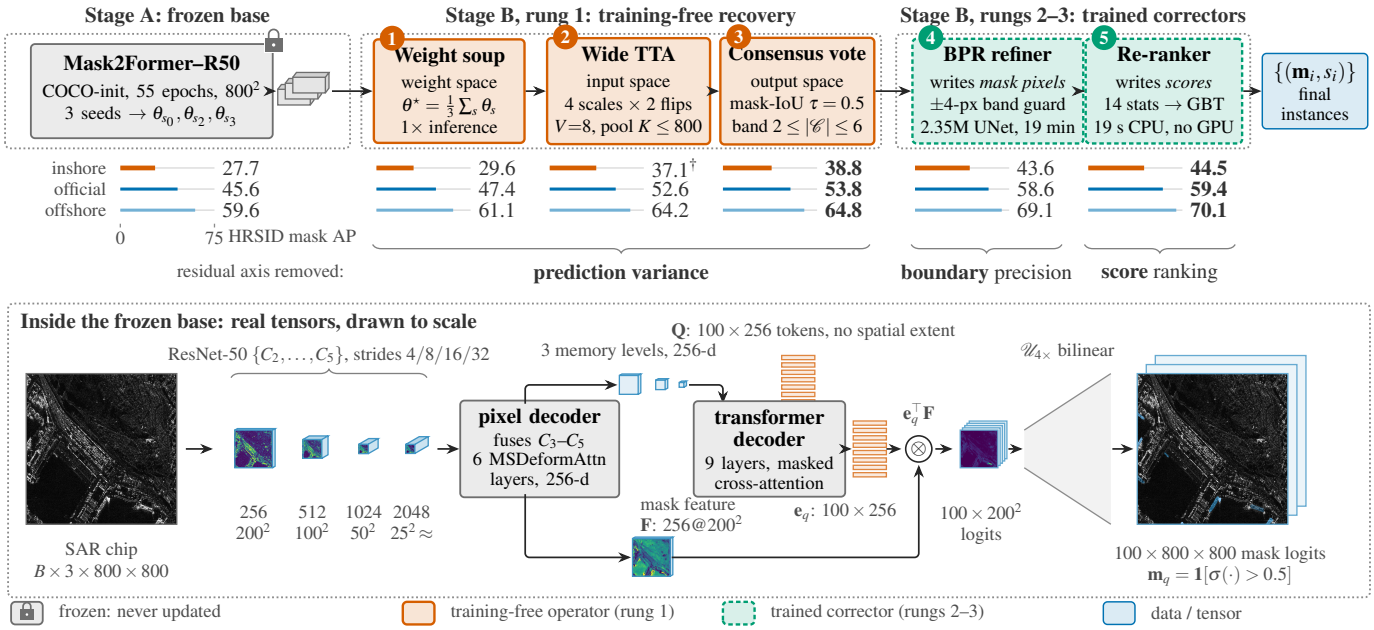

Fig. 1. Overall architecture of the proposed framework. Color key: gray with a padlock, never updated; vermillion, the three training-free operators of rung 1; dashed green, the two optional trained correctors of rungs 2–3; blue, data and tensors. The bar triple under each stage is inshore/official/offshore mask AP, and † marks the box-NMS merge that operator 3 replaces. The band underneath is one real forward pass on HRSID inshore chip `P0101_1200_2000_3000_3800.jpg` (image id 1528), in which every map shown is the actual dumped tensor.

constraint: no layer, loss, or training schedule of Stage A is modified. Section IV supports this constraint empirically with a seven-direction ablation in which every architecture- and training-side modification of this base was flat, negative, or absorbed.

C. Same-Basin Weight Soup

Fine-tuning is stochastic. Three independently seeded runs land within 0.1 inshore AP of one another (27.7/27.6/27.7), yet their prediction errors differ, and the first operator removes this seed variance in weight space. The three checkpoints are averaged elementwise into a uniform model soup [15],

$$\theta^\star = \frac{1}{3} \sum_{s \in \{s_0, s_2, s_3\}} \theta_s, \tag{3}$$

where θ_{s_0}, θ_{s_2}, and θ_{s_3} are the parameter vectors of the three fine-tuned checkpoints. Two same-basin conditions make the average a flat-minimum model rather than a broken interpolation: all members are fine-tuned from the same COCO initialization with identical parameter keys, and their batch-normalization buffers are byte-equal. Fig. 2 draws the measured geometry: over all 565 floating-point tensors (44.2 M parameters) of the three checkpoints the pairwise distances $\|\theta_{s_i} - \theta_{s_j}\|_2$ are 30.16, 29.81, and 30.18, so the seeds sit at almost exactly equal separations, and $\theta^\star$ is their centroid at 17.27, 17.49, and 17.29 from them; the released checkpoint reproduces the recomputed uniform mean to 1.6×10^{-7}, that is, to float32 rounding. These distances are recomputed by `figs/measure_soup_geometry.py` into `figs/assets/soup_geometry.json`. The soup is computed once, offline, and yields a single deterministic checkpoint at $1\times$ inference cost. It lifts the three-seed mean

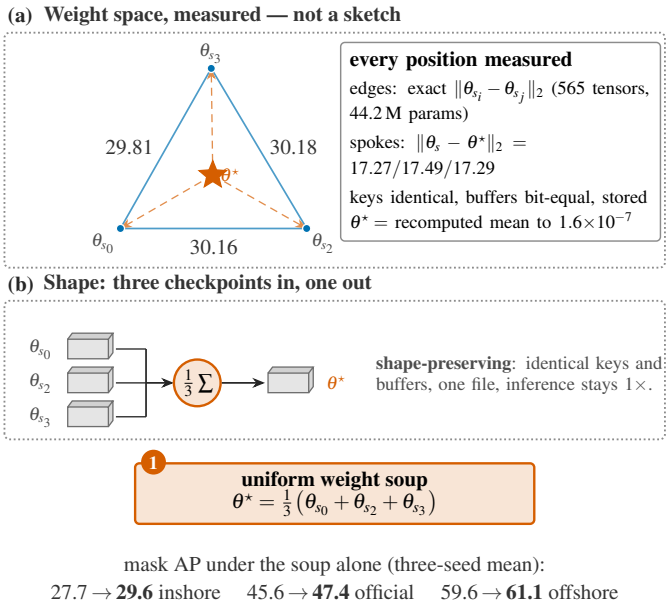

Fig. 2. Measured weight-space geometry of the same-basin soup. (a) The triangle is the exact two-dimensional embedding of the three measured pairwise distances $\|\theta_{s_i} - \theta_{s_j}\|_2$, and the star is the uniform elementwise average $\theta^\star$, which is the centroid of any Euclidean embedding of three points; dashed vermillion spokes are $\|\theta_s - \theta^\star\|_2$. No loss axis is drawn, because the figure reports measured geometry and not the loss surface. (b) Three checkpoints enter and one leaves, so inference stays $1\times$.

from 27.7 to 29.6 inshore (+1.9), 45.6 to 47.4 official, and 59.6 to 61.1 offshore. Under the full stack this single checkpoint matches a three-model prediction ensemble (39.3/53.6/64.0) at one third of its inference cost (Section IV).

D. Wide Multi-Scale Flip TTA

The second operator attacks scale-induced variance in input space. The souped model $\Phi_{\theta^\star}$ is run under the view set

$$\mathcal{V} = \{640,\ 800,\ 960,\ 1088\} \times \{\mathrm{id},\ \mathrm{hflip}\}, \qquad V = |\mathcal{V}| = 8, \tag{4}$$

where each pair $(\rho, f) \in \mathcal{V}$ resizes the chip to $\rho \times \rho$ and optionally applies a horizontal flip. The pooled candidate set is

$$\big\{(\mathbf{m}_k, s_k)\big\}_{k=1}^{K} = \bigcup_{(\rho,f)\in\mathcal{V}} f^{-1}\Big(\Phi_{\theta^\star}\big(f(R_\rho(\mathbf{x}))\big)\Big), \tag{5}$$

where R_ρ denotes bilinear resizing to $\rho\times\rho$, f^{-1} inverts the flip on the predicted masks (the identity for unflipped views), and the mask head rescales its outputs (`rescale=True`) so that every view returns masks at the native 800×800 resolution; consequently $K \le 8\times100 = 800$ scored candidates per image. Fig. 3 depicts this fan-out and pooling geometry.

The set is deliberately wide. A six-view core $\{736, 800, 864\} \times \{\mathrm{id}, \mathrm{hflip}\}$ already lifts single-checkpoint inshore AP by +7.0 on average (+6.9/+7.4/+6.7 across the three seeds, $\sigma \approx 0.3$), with a regime gradient (inshore +7.0, official +5.5, offshore +3.4) concentrated exactly where speckle and dock clutter are hardest. Widening to the eight views of Eq. (4) adds a further +2.0 inshore on seed s_0 (34.6 to 36.6) yet is flat on the official split and costs 0.6 offshore: the upsampled scales 960 and 1088 place more pixels on pixel-starved near-shore hulls, while large, well-resolved offshore ships gain nothing. The soup and TTA axes stack near-additively; their standalone inshore gains (+1.9 and +6.9) predict 36.5 where the measured combination reaches 36.1, a mildly sub-additive interaction, and we claim no synergy. What the pooled views do demand is a fusion rule. Under plain box-NMS merging the soup $\times$ 8-view configuration reaches 37.1/52.6/64.2 inshore/official/offshore; the next subsection shows why this reference fuser underuses the ensemble and how two plausible alternatives fail outright.

E. Consensus-Gated Mask-IoU Vote Fusion

Merging up to 800 overlapping candidates is where aggressive TTA either pays or collapses, and the observed failure modes dictate the design. Box-space suppression is geometrically wrong for harbors: two ships berthed in parallel produce boxes with high IoU but masks with low IoU, so box-NMS structurally deletes true neighbors. Score-decay suppression fails differently. SOLO-style matrix-NMS only attenuates the roughly eight near-duplicates that $V = 8$ views generate per object, floods the ranking with survivors, and collapses to 4.1/8.9/14.4 mask AP on the identical candidate set. Unconditional pixelwise averaging fails on the opposite flank: majority voting over every cluster erodes long, well-resolved offshore boundaries to 55.9, below the 59.6 no-TTA base, even though the same candidates reach 64.2 offshore under box-NMS. The fuser, not the views, is the safety condition. Our operator therefore clusters in mask-IoU space, deletes duplicates hard, and averages only where the ensemble itself signals disagreement.

All K candidate masks are flattened into rows of $\mathbf{M} \in \{0,1\}^{K\times HW}$, and the pairwise mask-IoU (Jaccard) matrix is

$$J_{jk} = \frac{\mathbf{m}_j^\top \mathbf{m}_k}{\mathbf{1}^\top\mathbf{m}_j + \mathbf{1}^\top\mathbf{m}_k - \mathbf{m}_j^\top\mathbf{m}_k}, \tag{6}$$

where $\mathbf{m}_j^\top\mathbf{m}_k$, the (j,k) entry of the Gram matrix $\mathbf{M}\mathbf{M}^\top$, counts intersection pixels and $\mathbf{1}^\top\mathbf{m}_j$ is the area of mask j. Candidates are consumed greedily in score order. With $\mathcal{U}$ the set of not-yet-consumed indices, one cluster is formed per iteration as

$$\mathcal{C} = \{r\} \cup \big\{j \in \mathcal{U} : c_j = c_r,\ J_{rj} \ge \tau\big\}, \qquad r = \arg\max_{k\in\mathcal{U}} s_k, \tag{7}$$

where r indexes the seed (the highest-scoring unconsumed candidate), c_j is the class label of candidate j, and $\tau = 0.5$ is the mask-IoU threshold. Each cluster is read out under a cross-view agreement gate,

$$\mathbf{m}^{\mathrm{out}}(p) = \begin{cases} \mathbf{1}\Big[\sum_{j\in\mathcal{C}} \mathbf{m}_j(p) > \frac{|\mathcal{C}|}{2}\Big], & 2 \le |\mathcal{C}| \le S_{\max}, \\ \mathbf{m}_r(p), & \text{otherwise}, \end{cases} \tag{8}$$

where p ranges over pixels and $S_{\max} = 6$ bounds the agreement band. The output score is

$$s^{\mathrm{out}} = s_r\,\big(|\mathcal{C}|/V\big)^\gamma, \qquad \gamma = 0, \tag{9}$$

where $\gamma = 0$ reduces the rule to plain seed-score inheritance; the general form is kept because the consensus ratio $|\mathcal{C}|/V$ is the quantity the gate consumes. The output box is regenerated from the fused mask. Algorithm 1 lists the complete procedure, and Fig. 4 shows the operator on a real candidate pool, together with the two failure modes that motivate its gate. The clusters it produces are disjoint in practice: on the eight-view pool of one inshore chip the 16 highest-scoring candidates fall into four clusters of sizes 6, 7, 1, and 2, and every off-block entry of the mask-IoU matrix, recorded in `figs/assets/iou_matrix_meta.json`, is exactly zero.

The agreement band reads the ensemble's own cross-view consensus as a per-instance variance estimate. A near-unanimous cluster ($|\mathcal{C}| > 6$) is a clean detection whose seed mask any averaging can only erode; it is emitted verbatim. A singleton has nothing to vote over and is likewise preserved. Clusters inside the band exhibit genuine cross-view disagreement, typically ambiguous inshore ships, and only these are majority-denoised. The gate is monotone in the band ceiling (inshore 38.4/38.6/38.8 for $S_{\max} = 4/5/6$, with offshore fixed at 64.8), and the full operator reaches 38.8/53.8/64.8, matching or exceeding the seed-only readout (38.2/53.5/64.8) on all three splits. Composed, the stack lifts the 27.7/45.6/59.6 base by +11.1/+8.2/+5.2 with no weight updated.

F. Optional Boundary-Patch Refinement (BPR)

The stack recovers detection and recall, not boundary precision. Mask logits predicted at stride 4 (200×200) are upsampled $4\times$ bilinearly in Eq. (2), so predicted contours carry a soft halo of roughly one to two pixels, and on a typical 9×18-pixel inshore hull a one-pixel boundary shift already

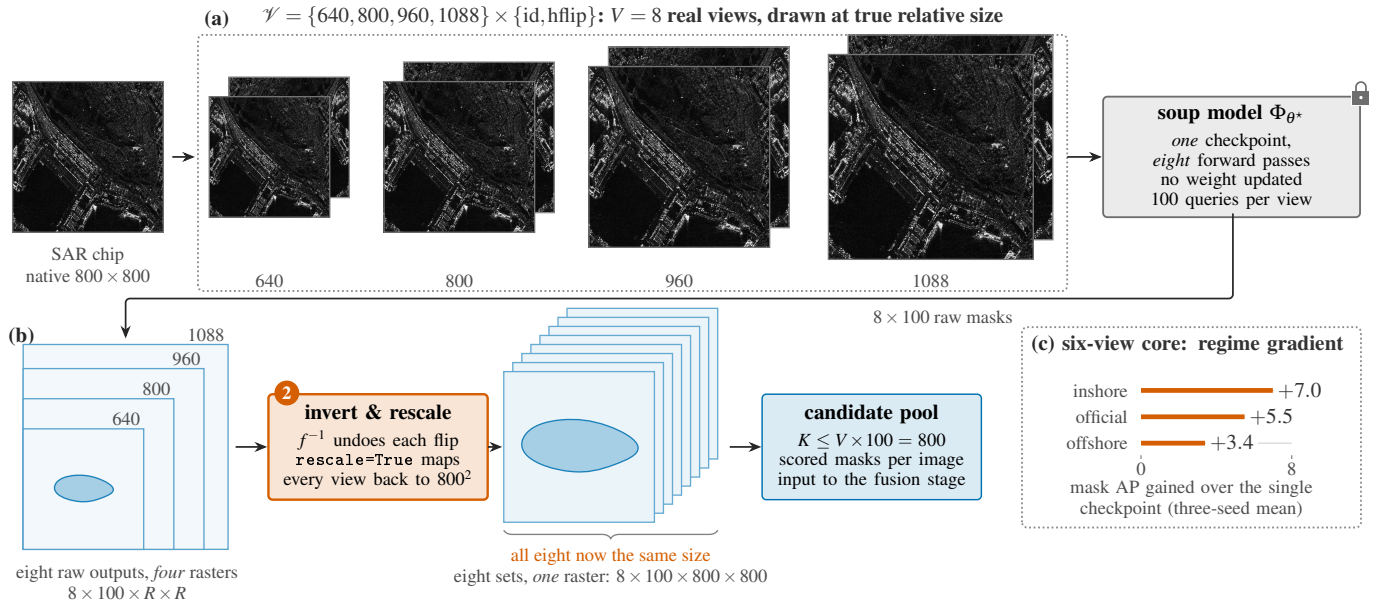

Fig. 3. Wide multi-scale flip TTA. (a) The eight views of Eq. (4) are the real rasters computed on HRSID inshore chip `P0101_1200_2000_3000_3800.jpg`, drawn at their true relative size of 16.0/20.0/24.0/27.2 mm, so the 1088 view places 2.9× the pixels of the 640 view on the same hull. (b) The same eight outputs, first on their four native rasters and then on one common 800 × 800 raster after f^{-1} and rescaling. (c) Measured mask AP gained by the six-view core over the single checkpoint, three-seed mean.

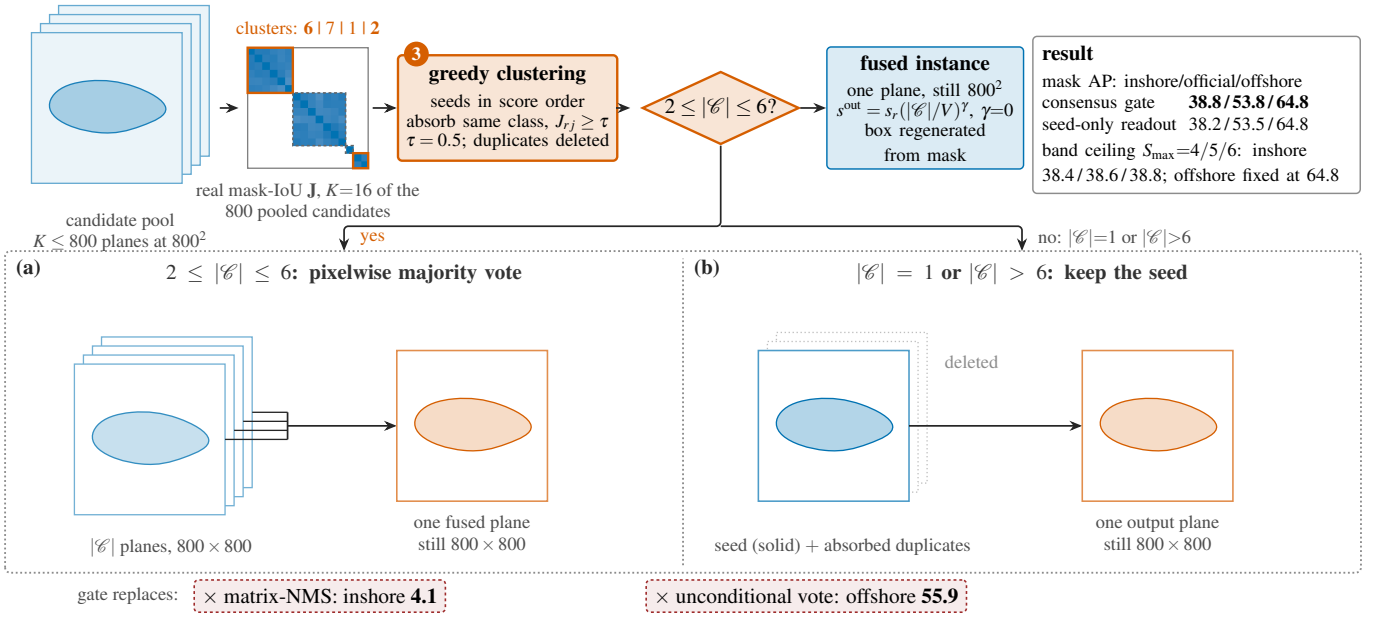

Fig. 4. Consensus-gated vote fusion on a real candidate pool. The matrix is the measured pairwise mask IoU of the 16 highest-scoring candidates of HRSID inshore chip `P0101_1200_2000_3000_3800.jpg`, permuted into greedy-cluster order; white is IoU 0, saturated blue IoU 1. Solid vermillion boxes mark the two clusters inside the agreement band, dotted gray boxes the two outside it: (a) in-band clusters collapse by pixelwise majority; (b) out-of-band clusters emit the seed plane. The two chips at the foot are the matrix-NMS and unconditional-voting fusers that the gate replaces.

costs about 0.15–0.25 IoU. No training-free control we tested moves this residual (Section IV). We therefore add an optional trained second rung in the BPR paradigm [18], constructed so that it cannot alter anything except boundary pixels; Fig. 5 summarizes its training and band-guarded application paths.

The refiner is trained once, in about 19 minutes. The full stack is run on the training split and its predictions are matched to training ground truth at mask IoU ≥ 0.5; square boundary patches of side 64, upscaled 2×, are cropped along each matched contour, forming (prediction, ground-truth) patch pairs. A 2.35M-parameter UNet is trained on these pairs with

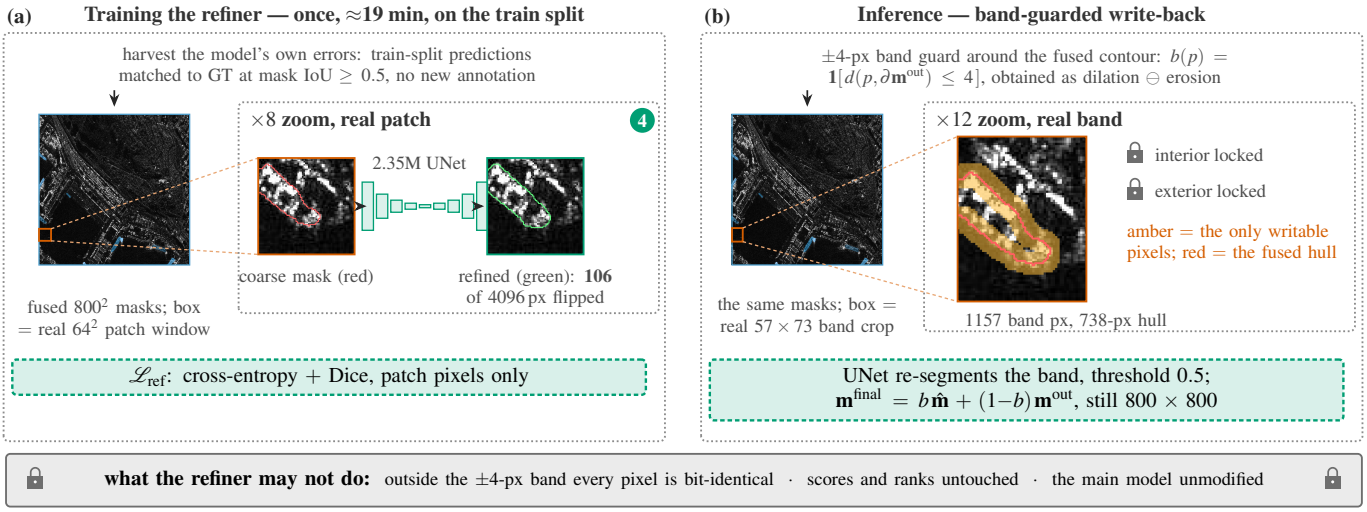

Fig. 5. Boundary-patch refinement on the real fused instance masks of HRSID inshore chip `P0101_1200_2000_3000_3800.jpg`. (a) Training: the vermillion box is the real patch window at origin $(536, 0)$, 1.6 mm against the 20 mm mask and hence shown at $\times 8$; inside the zoom it appears before and after refinement, red contour coarse and green refined. (b) Inference: the vermillion box is the real 57×73 px band crop and the $\times 12$ zoom shows the annulus $\mathrm{dilate}(m, 4) \ominus \mathrm{erode}(m, 4)$, in which amber is the only editable region; padlocks mark its interior and its exterior.

Algorithm 1 Consensus-Gated Mask-IoU Vote Fusion

Require: candidates $\{(\mathbf{m}_k, s_k)\}_{k=1}^{K}$ from $V = 8$ views; mask-IoU threshold $\tau = 0.5$; agreement band $[2, S_{\max}]$ with $S_{\max} = 6$; $\gamma = 0$
Ensure: fused instance set $\mathcal{Y}$

```
sort candidates by score, descending; mark all unused; Y ← ∅
while unused candidates remain do
    r ← highest-scoring unused candidate; C ← {r}
    absorb every unused same-class j with J_rj ≥ τ into C
    mark C used
    if 2 ≤ |C| ≤ S_max then
        m^out ← pixelwise majority vote over C {denoise}
    else
        m^out ← m_r {singleton or near-unanimous: preserve seed verbatim}
    end if
    s^out ← s_r (|C|/V)^γ; box ← mask2bbox(m^out)
    append (m^out, s^out) to Y
end while
return top-100 of Y by score
```

a combined binary cross-entropy and Dice objective,

$$\mathcal{L}_{\mathrm{ref}} = -\frac{1}{|\Omega|}\sum_{p\in\Omega}\left[y_p \log \hat{y}_p + (1-y_p)\log(1-\hat{y}_p)\right] + 1 - \frac{2\sum_{p\in\Omega}\hat{y}_p\, y_p + \varepsilon}{\sum_{p\in\Omega}\hat{y}_p + \sum_{p\in\Omega} y_p + \varepsilon}, \tag{10}$$

where Ω is the set of patch pixels, $\hat{y}_p \in (0, 1)$ the refiner's sigmoid output at pixel p, $y_p \in \{0, 1\}$ the ground-truth label, and ε a small stability constant. The main model is never touched, and the filename overlap between refiner training patches and test images is zero.

Application is band-guarded. At test time the refiner re-segments the boundary patches of each detected instance, and the refined prediction is written back only inside a ± 4-pixel band around the original boundary,

$$\mathbf{m}^{\mathrm{final}}(p) = b(p)\,\hat{\mathbf{m}}(p) + \big(1 - b(p)\big)\,\mathbf{m}^{\mathrm{out}}(p), \qquad b(p) = \mathbf{1}\big[d\big(p, \partial\mathbf{m}^{\mathrm{out}}\big) \le 4\big], \tag{11}$$

where $\hat{\mathbf{m}}$ is the refiner output stitched over the patches, $\partial\mathbf{m}^{\mathrm{out}}$ the boundary of the stack's mask, and $d(\cdot, \cdot)$ the pixel distance to that boundary. Overlapping patch logits are averaged inside the band and thresholded at 0.5. The resulting edit is small as well as confined: in the instance drawn in Fig. 5, refinement flips 106 of the 4096 pixels of one boundary patch, and the ± 4-pixel band around a 738-pixel hull holds 1157 editable pixels. Every pixel outside the band is bit-identical to the stack's output, and detection scores, ranks, and instance identities are untouched: the refiner can displace boundaries by a few pixels but can neither add, remove, nor re-rank detections. With this rung the pipeline reaches 43.6/58.6/69.1 mask AP and 47.0/70.7/87.4 AP_{75} on inshore/official/offshore. The metric fingerprint is that of a pure boundary correction, since inshore AP_{50} moves by only +0.7 while AP_{75} moves by +10.9; Section IV audits the application path along seven leakage vectors. The training-free claim of this article is accordingly scoped to the stack of Secs. III-C–III-E, and the refiner, like the re-ranker that follows, is reported separately in every table.

G. Learned Consensus Re-Ranking

Two axes of the residual are now treated and a third is not. Mask AP is a ranking metric, so a candidate whose mask is

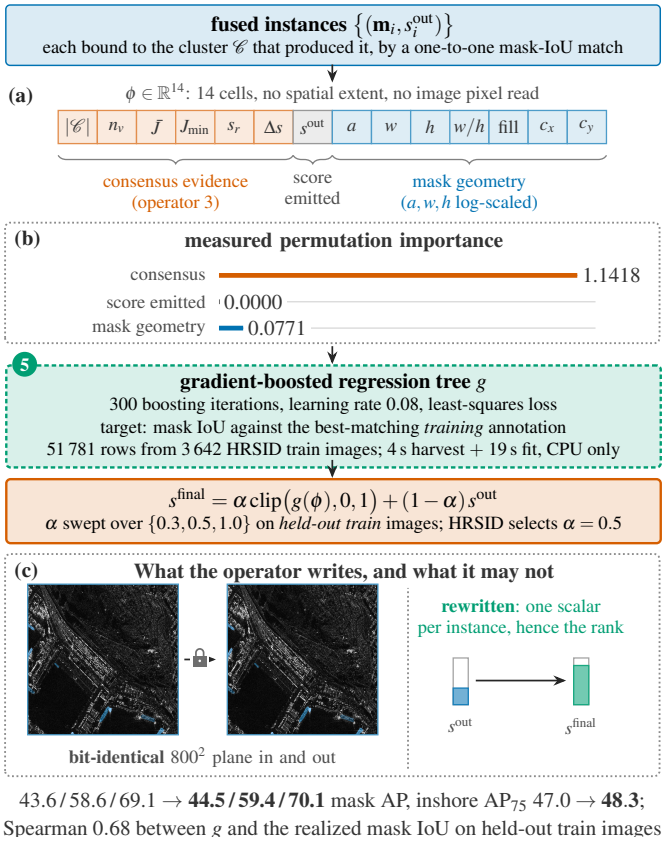

Fig. 6. The learned consensus re-ranker. (a) Each fused instance is described by the 14 scalars of Eqs. (12)–(13): six consensus statistics (vermillion), the score the fusion stage emits (gray), and seven mask-geometry descriptors (blue). (b) Measured permutation importance of the three feature groups. (c) Only the score field is rewritten: the two instance-mask rasters are byte-identical files, and only the height of the score bar changes.

accurate but whose score falls below a worse neighbor forfeits precision at every recall level it should have dominated. The fusion operator of Sec. III-E already computes the quantity that predicts mask quality, the cross-view consensus of each cluster, and then discards it: Eq. (9) reduces to seed-score inheritance at $\gamma = 0$ because no fixed exponent on $|\mathcal{C}|/V$ improved all three splits at once. Rather than search harder for a formula, we stop writing one. The base model makes its errors on the training split as readily as on the test split, and those errors are free supervision, so the map from consensus evidence to mask quality is fitted instead of guessed. Fig. 6 summarizes the operator.

Each fused instance is described by a 14-dimensional vector of scalars,

$$\phi = \left[\phi_{\text{c}}^{\top},\ s^{\text{out}},\ \phi_{\text{g}}^{\top}\right]^{\top} \in \mathbb{R}^{14}, \tag{12}$$

whose consensus and geometry blocks are

$$\begin{aligned}\phi_{\text{c}} &= \left[\,|\mathcal{C}|,\ n_v,\ \bar{J},\ J_{\min},\ s_r,\ \Delta s\,\right]^{\top},\\ \phi_{\text{g}} &= \left[\log(1{+}a),\ \log(1{+}w),\ \log(1{+}h),\right.\\ &\qquad \left. w/h,\ a/(wh),\ c_x,\ c_y\,\right]^{\top},\end{aligned} \tag{13}$$

where $|\mathcal{C}|$ is the cluster size of Eq. (7), n_v the number of distinct views of Eq. (4) contributing to that cluster, $\bar{J}$ and $J_{\min}$ the mean and the minimum pairwise mask IoU of Eq. (6) inside it, s_r the seed score, Δs the spread between the largest and the smallest member score, s^{out} the score the fusion stage emits, a the mask area, w and h the width and height of the mask box, and (c_x, c_y) the box center normalized by image width and height. Under the deployed setting $\gamma = 0$ the two score channels s_r and s^{out} coincide, which costs a tree ensemble nothing and keeps the same extractor valid for $\gamma > 0$. No image pixel is read anywhere in Eq. (13): every entry is a by-product of computation the stack has already performed, which is what makes the operator cost microseconds per instance.

The regression target is the mask IoU between the candidate and its best-matching training annotation, and zero when the candidate overlaps none. Every candidate above a score floor of 0.01 contributes a row, kept and rejected alike, so the model sees the failures and not only the survivors; HRSID yields 51,781 rows from 3,642 training images and PSeg-SSDD 5,122 rows from 928. A histogram-based gradient-boosted regression tree g (300 boosting iterations, learning rate 0.08, least-squares loss) is fitted to these rows, and its prediction is blended into the detection score as

$$s^{\text{final}} = \alpha\,\text{clip}\big(g(\boldsymbol{\phi}), 0, 1\big) + (1-\alpha)\,s^{\text{out}}, \tag{14}$$

where $\alpha \in [0, 1]$ trades the learned quality estimate against the network's own confidence and the clip keeps the regressor inside the score range.

The mixing weight is the only hyperparameter, and it is selected without touching test data. Training images are partitioned by image identifier into a fit part and a holdout part in a 4:1 ratio (2,914 and 728 images on HRSID), g is fitted on the fit part alone, and α is swept over $\{0.3, 0.5, 1.0\}$ by COCO segm AP measured on the held-out training images. HRSID selects $\alpha = 0.5$, which moves that train-side proxy from 57.6 to 58.9 AP; PSeg-SSDD, whose re-ranker is refitted from scratch on its own training split, selects $\alpha = 1.0$ and moves its proxy from 65.0 to 67.8. The chosen weight is then frozen and g is refitted on all training rows. Everything runs on a CPU: 4 s to harvest features and 19 s to fit on HRSID, 2 s and 9 s on PSeg-SSDD. On the held-out training images the Spearman correlation between g and the true mask IoU is 0.68 on HRSID and 0.86 on PSeg-SSDD, and permutation importance ranks the score channel first by a wide margin, then mask area, fill ratio, and the view count n_v. Grouped by feature block over the held-out 20% of the 3 642 HRSID training images, with three repeats, that importance reads 1.1418 for the six consensus statistics, 0.0771 for the seven geometry descriptors, and 0.0000 for the emitted score, the seed score alone accounting for 1.0944 of the consensus total (Fig. 6); the measurement is reproduced by `figs/measure_rerank_importance.py`. The operator does not replace the network's confidence; it corrects it with evidence the network never saw in a single forward pass.

Two safety properties follow from what the operator is allowed to write. First, the candidate set is preserved exactly. Masks, instance identities, and the size of the output set are bit-identical to the input, and only the score field changes, so re-ranking can neither create, delete, nor reshape a detection.

Each detection of the preceding rung is bound to its fusion-stage candidate by a greedy one-to-one mask-IoU match at threshold 0.5, purely to retrieve that candidate's consensus statistics, and a detection with no match keeps its original score untouched; on the HRSID inshore split 23,857 of 28,539 detections are matched. Second, the refiner and the re-ranker write to disjoint fields. The refiner edits mask pixels inside a ± 4-pixel band, the re-ranker reads scalars produced by the fusion stage and edits scores, so neither can undo the other and the order of application is a matter of convenience; we apply the refiner first so that the masks finally reported are the refined ones. On HRSID the re-ranking rung lifts 43.6/58.6/69.1 to 44.5/59.4/70.1 mask AP and inshore AP_{75} from 47.0 to 48.3; on PSeg-SSDD it lifts 60.0/66.3/68.8 to 62.1/67.5/69.8. A carrier or a dataset new to the recipe needs its own g, since the consensus statistics belong to that carrier, and refitting one costs under a CPU-minute and no GPU time.

H. Complexity

The soup costs one offline elementwise average and nothing at inference; the recovery stack updates no weights and adds no parameters to the base. Test-time cost is dominated by the $V = 8$ forward passes of Eq. (5), and the measured end-to-end cost of the full method is approximately $8\times$ a single-view pass. Fusion adds one $K \times K$ Gram product over binary masks and a greedy scan with $K \leq 800$; this overhead is contained in the measured $8\times$ figure. The optional refiner adds a 2.35M-parameter UNet, trained once in about 19 minutes, and a per-instance boundary-patch pass at test time; total cost remains approximately $8\times$ plus this small constant. The optional re-ranker is cheaper still. Its features are already computed by the fusion stage, one gradient-boosted tree evaluation over 14 scalars costs microseconds per instance, and no image is read a second time, so the rung is free to the nearest measurable unit at test time and costs 23 s of CPU once at fit time on HRSID (11 s on PSeg-SSDD). Stage A itself trains for 55 epochs on one GPU, against the 500-epoch specialized protocol of [2]. No component of Stage B touches a single weight of Stage A, so the recipe wraps any Mask2Former-class base unchanged.

IV. Experiments

This section presents the experimental evaluation on HRSID and a cross-dataset transfer to PSeg-SSDD. We first describe the dataset, the training protocol, and the seed discipline. We then report the progressive main results, compare with published state-of-the-art methods, and present two controlled ablations: the fusion operator and the architecture side. We next analyze the two optional trained tiers, the boundary-patch refiner and the learned re-ranker, each with its own leakage audit, and examine qualitative results on representative scenes, first against our own base and then against two detectors that we trained ourselves on the same split. We then transfer the complete recipe to PSeg-SSDD, compare it there against six methods that we trained ourselves under one protocol, and apply the identical recipe to six frozen carriers spanning five architecture families in order to test whether any of it depends on our own base. Finally, we re-read the same predictions as a detection-level false-alarm and missed-detection accounting and verify the robustness of the recovery gain across seeds.

A. Dataset and Protocol

Dataset and splits. All experiments were conducted on HRSID, evaluated as SAR ship instance segmentation; every number in this section is COCO mask AP $\times 100$, reported to one decimal. The dataset comprises 5604 SAR chips of 800×800 pixels with instance-level mask annotations, split into 3642 training and 1962 test images. Following the inshore/offshore protocol of prior work [2], the official test set is additionally partitioned into an inshore subset of 369 images and an offshore subset of 1593 images. We report all three splits throughout: the official split has no published competitor row and is given for completeness, whereas the inshore split is where the open problem lives.

Training protocol. The base model, denoted c0, is a standard Mask2Former (M2F) [7] with a ResNet-50 backbone and 100 queries, initialized from COCO instance-segmentation weights. It was fine-tuned at 800×800 with AdamW at an initial learning rate of 2.5×10^{-5} with step decay, batch size 2 with 2-step gradient accumulation (effective batch 4), for 100 872 iterations; at two images per iteration this amounts to 201 744 image passes over the 3642-image training split, i.e., 55 epochs. Training ran on a single NVIDIA RTX 3090 Ti. No layer, loss, or schedule of the base was modified.

Seed discipline and evaluation. All base-level claims use three independent seeds (s0/s2/s3); the inter-seed standard deviation of the base is approximately ± 0.3 AP, and single-seed differences within this band are treated as noise. The weight soup is a deterministic single checkpoint (the elementwise average of the three seeds), and the TTA evaluation is deterministic inference, so every stack-level figure is reproducible from fixed artifacts. All of our configurations were scored against the identical official ground truth with the identical COCO evaluator; every within-article delta is therefore a same-GT, same-protocol comparison. Competitor numbers in Section IV-C are taken from their published tables and were not re-evaluated by us; the attendant caveats are stated there.

B. Main Results

Table I and Fig. 7 report the progressive recovery stack. Each row adds one operator to the previous row. No row retrains or modifies the base. The weight soup lifts the three-seed base from 27.7/45.6/59.6 to 29.6/47.4/61.1 inshore/official/offshore at unchanged $1\times$ inference; six-view multi-scale flip-TTA raises the inshore split to 36.1; widening to eight views over $\{640, 800, 960, 1088\} \times \{\text{identity}, \text{flip}\}$ yields 37.1/52.6/64.2 under box-NMS; and replacing box-NMS with the consensus-vote fusion of Section III-E completes the full method at **38.8/53.8/64.8**. The composed lift over the base is +11.1/+8.2/+5.2, a relative inshore gain of +40%, at zero additional training and roughly $8\times$ inference.

The first two operators compose near-additively. Soup alone contributes +1.9 inshore and the six-view TTA alone +6.9, predicting 36.5 where the measured combination is 36.1,

TABLE I
PROGRESSIVE TRAINING-FREE RECOVERY ON HRSID (MASK AP $\times 100$). EACH ROW ADDS ONE OPERATOR TO THE PREVIOUS ROW; NO ROW RETRAINS OR MODIFIES THE BASE. FULL COCO METRICS ARE REPORTED FOR THE THREE AUDITED ENDPOINTS

			Inshore			Official (full)			Offshore		
Configuration	Train. cost	Infer. cost	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
c0 base, COCO-init M2F-R50 (3-seed mean)	55 ep	$1\times$	27.7	61.8	22.1	45.6	81.9	49.4	59.6	97.2	71.2
+ weight soup (avg. of s0/s2/s3)	0	$1\times$	29.6	–	–	47.4	–	–	61.1	–	–
+ 6-view msTTA $\{736, 800, 864\}\times$flip	0	$\sim 6\times$	36.1	–	–	52.5	–	–	64.4	–	–
+ wide 8-view TTA $\{640, 800, 960, 1088\}\times$flip, box-NMS	0	$\sim 8\times$	37.1	–	–	52.6	–	–	64.2	–	–
+ consensus-vote fusion (**full method**)	0	$\sim 8\times$	**38.8**	**75.6**	**36.1**	**53.8**	**89.4**	**62.2**	**64.8**	**98.5**	**81.7**
+ BPR refiner (optional trained tier, Sec. IV-F)	19 min	$\sim 8\times + \varepsilon$	43.6	76.3	47.0	58.6	89.9	70.7	69.1	98.6	87.4
+ learned re-ranking (optional trained tier, Sec. IV-G)	19 s	$\sim 8\times + \varepsilon$	44.5	76.7	48.3	59.4	90.1	71.4	70.1	98.6	88.0

Bold denotes the training-free full method. "–": full metrics are logged only for the three audited endpoints.
The two lower rows are optional trained tiers and are reported separately from the training-free tier: the refiner is a 2.35M-parameter post-processor trained for about 19 minutes, and the re-ranker is a 14-feature gradient-boosted regressor fitted in 19 seconds on CPU that re-orders the same detections without editing a single mask pixel.

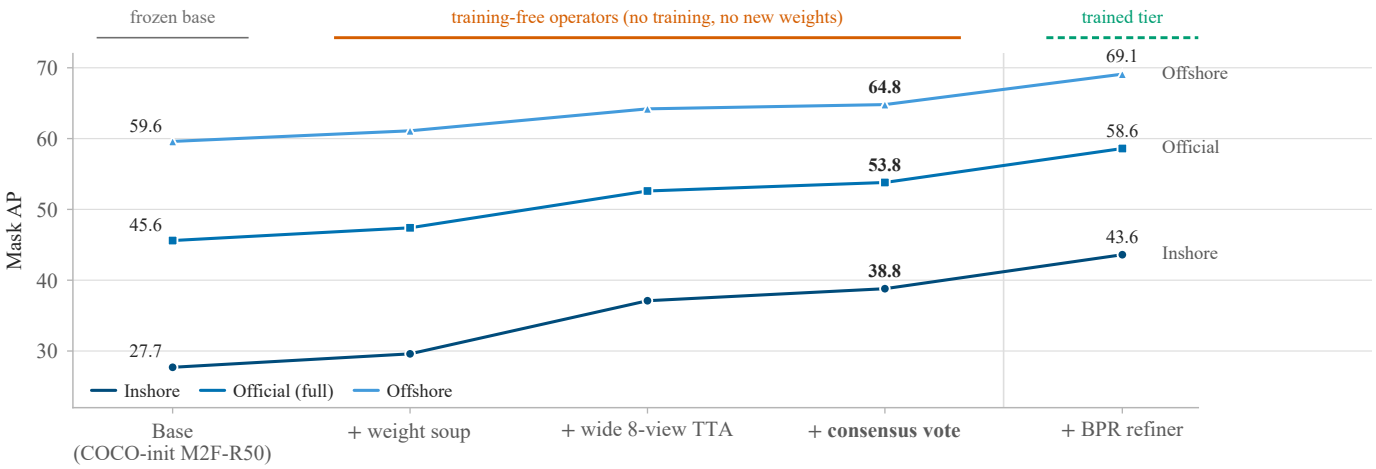

Fig. 7. Progressive training-free recovery on HRSID (Table I). Each trajectory is one test split read across the rungs of the recovery ladder; the darkest is inshore and the lightest offshore. The first four rungs are training-free and the last is the optional trained BPR tier. The composed training-free lift is +11.1/+8.2/+5.2 over the c0 base; with the refiner it becomes +15.9/+13.0/+9.5, and with the learned re-ranker of Section IV-G, tabulated in Table I but not plotted here, +16.8/+13.8/+10.5.

a -0.4 sub-additive interaction; we explicitly do not claim synergy. The fusion operator then contributes +1.7/+1.2/+0.6 over box-NMS on the identical candidate set; Section IV-D shows this term is an enabler rather than a third additive gain. On the full-metric axis, the stack moves inshore AP_{50} from 61.8 to 75.6 (+13.8), inshore AP_{75} from 22.1 to 36.1 (+14.0), and inshore AP_s from 24.3 to 35.8, so detection, strict-overlap quality, and small-target coverage all participate in the recovery. As a control, a three-model prediction ensemble over the same eight views reaches 39.3/53.6/64.0 at $3\times$ model cost against 38.8/53.8/64.8 for the soup-based method at $1\times$, confirming that the weight soup captures the seed diversity of a full prediction ensemble at one third of its inference cost [15].

C. Comparison with State of the Art

Tables II and III compare against the published HRSID inshore and offshore results under the same split protocol. Competitor rows are as published: the classic-detector and from-scratch Mask2Former rows are tabulated by O2Former [2], the Gaussian-heatmap row is from its TGRS article [4], and the DiffSARShipInst row is relayed through O2Former's secondary table because its primary source is paywalled [3]; we note that its offshore AP_{50} of 97.8 and AP_{75} of 90.0 have, to our knowledge, not been independently reproduced.

Two results in Table II carry the comparison. With both optional tiers, inshore reaches 44.5 AP, 48.3 AP_{75}, and 42.0 AP_s, leading every published method in all three columns by +1.4, +2.4, and +2.1 over O2Former, the strongest of them, which reports 43.1, 45.9, and 39.9. That margin costs 55 epochs plus a 19-minute post-processor and a 19-second re-ranker, against 500 epochs of specialized training. The refiner alone already reads 43.6/47.0/41.4 and leads the same three columns, so the leadership does not hinge on the re-ranking tier. The training-free stack alone already posts an inshore AP_{50} of 75.6, exceeding O2Former's 69.5 by +6.1, although it does not lead on absolute inshore AP (38.8 against 43.1) and its AP_{75} of 36.1 trails their 45.9: the stack is a detection-and-recall recovery, not a boundary-precision method. Offshore, the complete pipeline trails the two strongest specialized methods (70.1 against 71.7 and 70.9), with the training-free stack further behind on that split. Even there, our offshore AP_{50} of 98.6 is the highest in Table III, and the near-shore

TABLE II
COMPARISON ON THE HRSID INSHORE SPLIT (MASK AP $\times 100$). COMPETITOR ROWS AS PUBLISHED; "SCRATCH" DENOTES AN IMAGENET BACKBONE WITH A FROM-SCRATCH SEGMENTATION HEAD; "N/R" DENOTES NOT REPORTED

Method	Backbone	Epochs	Init	AP	AP_{50}	AP_{75}	AP_s
Ours, full method + BPR + re-ranking	R-50	55 (+19 min)	COCO	**44.5**	76.7	**48.3**	**42.0**
Ours, full method + BPR	R-50	55 (+19 min)	COCO	43.6	76.3	47.0	41.4
O2Former† [2]	R-50	500	scratch	43.1	69.5	45.9	39.9
DiffSARShipInst‡ [3]	n/r	n/r	n/r	42.6	76.1	44.1	40.3
Gaussian-Heatmap§ [4]	n/r	n/r	n/r	42.5	**79.1**	39.9	41.0
RTMDet†	R-50	500	scratch	40.4	75.9	38.7	39.3
HQ-ISNet†	R-50	500	scratch	39.6	72.5	41.0	39.0
Ours, full method (training-free)	R-50	55	COCO	38.8	75.6	36.1	35.8
Cascade Mask R-CNN†	R-50	500	scratch	35.6	59.1	40.1	32.6
Mask R-CNN†	R-50	500	scratch	34.0	56.5	38.3	31.0
Ours, c0 base	R-50	55	COCO	27.7	61.8	22.1	24.3
Mask2Former† (their baseline)	R-50	500	scratch	17.9	36.9	16.2	14.7

Bold denotes the best value in each column. The two trained tiers add 19 minutes (refiner) and 19 seconds (re-ranker) of training. †As tabulated by O2Former [2], Table 3; not re-evaluated by us.
‡Primary source paywalled; figures relayed via O2Former's table, cross-checked against the article abstract. §From [4], Table II; protocol not reported.

TABLE III
COMPARISON ON THE HRSID OFFSHORE SPLIT (MASK AP $\times 100$). PROVENANCE AS IN TABLE II; "BOTH TIERS" DENOTES THE BOUNDARY REFINER FOLLOWED BY THE LEARNED RE-RANKER

Method	AP	AP_{50}	AP_{75}	AP_s
O2Former	**71.7**	97.5	**91.0**	**71.1**
DiffSARShipInst	70.9	97.8	90.0	70.3
Ours, full method + both tiers	70.1	**98.6**	88.0	69.4
Gaussian-Heatmap	68.0	97.7	85.1	67.4
HQ-ISNet	67.9	97.5	87.1	66.5
RTMDet	67.4	97.7	83.7	66.8
Cascade Mask R-CNN	67.4	96.8	85.7	66.7
Mask R-CNN	65.3	95.8	83.4	64.5
Ours, full method (training-free)	64.8	98.5	81.7	64.1
Mask2Former (their baseline)	60.6	95.2	74.9	59.5
Ours, c0 base	59.6	97.2	71.2	58.6

split, where the open problem lives, is where the proposed pipeline leads every published method on AP, AP_{75}, and AP_s simultaneously. No competitor reports the official split; our figures there (59.4 with both tiers, 53.8 training-free, 45.6 base) are provided for future comparison.

One caveat dominates all others and we state it plainly: the two columns of results are not at the same operating point. The competitor protocol trains segmentation heads from scratch for 500 epochs on an ImageNet backbone, whereas we fine-tune a COCO-initialized model for 55 epochs. The same Mask2Former architecture scores 17.9 inshore under their protocol and 27.7 under ours, a 9.8-point separation before any contribution of this article. Absolute AP across the two regimes is therefore not directly comparable. This is mainly because the pretraining regime, not the architecture, sets the baseline level, which is itself a central finding of Section IV-E. The defensible claims are the training-free recovery (+11.1/+8.2/+5.2 on a strong base at zero extra training) and the two audited micro-tiers stacked on top of it, a band-guarded boundary correction (Section IV-F) and a mask-preserving re-ranking (Section IV-G). Together they spend a deliberation budget where the competitor protocol spends a training budget.

TABLE IV
FUSION OPERATORS ON IDENTICAL SOUP $\times$ 8-VIEW CANDIDATES (MASK AP $\times 100$)

Fusion operator	Inshore	Official	Offshore
box-NMS (reference)	37.1	52.6	64.2
matrix-NMS	4.1	8.9	14.4
majority pixel vote	**39.1**	48.7	55.9
probability mean	**39.1**	48.7	55.9
seed vote	38.2	53.5	**64.8**
consensus vote ($S_{\max} = 6$)	38.8	**53.8**	**64.8**

Bold denotes the best value in each column.

D. Fusion Operator Ablation

All operators in Table IV consume the identical soup $\times$ wide-8-view candidate set; only the merge rule changes. Fig. 8 visualizes the outcome. It can be observed from Table IV that the fuser, not the view count, decides whether aggressive TTA helps or harms.

Each failure has a distinct mechanism. Box-NMS suppresses in box-IoU space, and berthed parallel hulls share high box IoU while their masks barely overlap; hard deletion therefore removes true inshore neighbors, capping the reference at 37.1. Matrix-NMS is the collapse mode: its soft score decay deletes nothing, so with eight views each object arrives roughly eight times and the decayed near-duplicates flood the top-100 ranking, driving AP to 4.1/8.9/14.4. Majority pixel voting is the strongest inshore reading at 39.1 but erodes offshore to 55.9, below the 59.6 no-TTA base: multi-scale views misalign by one to two pixels along long hull boundaries, and unconditional pixelwise voting trims every disputed pixel. The probability-mean control replaces binary voting with floating-point averaging before thresholding and reproduces the majority result bitwise (39.1/48.7/55.9); the erosion therefore stems from cross-view misalignment, not from binarization loss, and the correct degree of freedom is which mask to keep, not how to average pixels. The seed vote keeps each cluster's highest-scoring mask verbatim: it never erodes (offshore 64.8) and beats box-NMS on all three splits (+1.1/+0.9/+0.6), but concedes inshore at 38.2 by forgoing

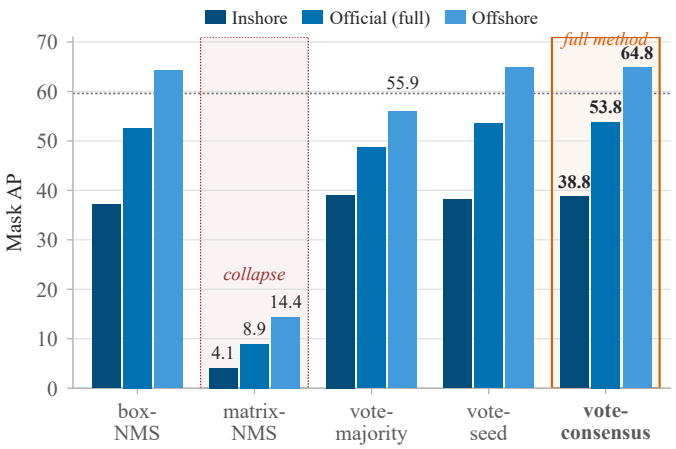

Fig. 8. Fusion-operator ablation on identical 8-view candidates (Table IV). Bar shade encodes the split, darkest inshore to lightest offshore. Matrix-NMS, in the dotted region, collapses to 4.1/8.9/14.4; majority voting erodes the offshore split to 55.9, below the dotted 59.6 no-TTA base; the consensus vote, in the solid region, matches or exceeds the seed vote on all three splits. The probability-mean control is not drawn as a sixth group because it reproduces the majority column bitwise (39.1/48.7/55.9).

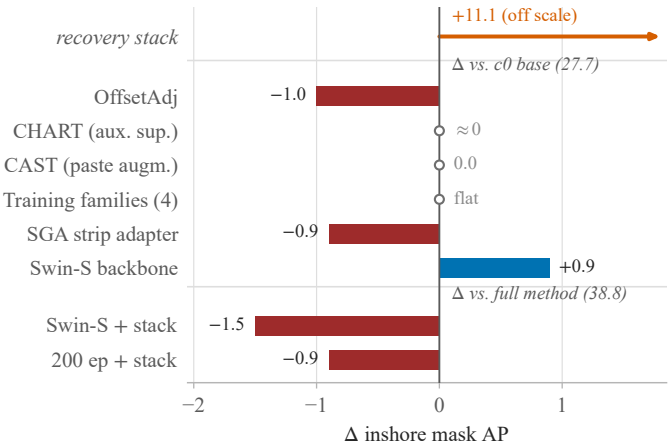

Fig. 9. Inshore mask-AP deltas of the architecture- and training-side probes (Table V). The upper group is measured against the c0 base (27.7) and the lower group against the full method (38.8); a hollow marker on the zero line denotes a flat direction. No probe reaches the training-free recovery, whose +11.1 runs off this scale and is drawn as the arrow at the top.

TABLE V
ARCHITECTURE- AND TRAINING-SIDE CONTROLS (MASK AP ×100). UPPER BLOCK: VS. THE C0 BASE; LOWER BLOCK: VS. THE FULL METHOD

Configuration	Inshore	Official	Offshore
c0 base (3-seed reference)*	27.7	45.7	59.8
OffsetAdj (constrained offsets)	26.7	45.3	59.7
CHART (aux. supervision, w=0.1)	≈27.7	45.7	59.8
CAST (hard-sample paste augm.)	27.7	–	–
Training-side families (4)	flat (subsumed by base)		
SGA (zero-gated strip adapter)	26.8	45.5	60.4
Swin-S backbone swap	28.6	46.1	59.9
Swin-S + recovery stack (1 seed)	37.3	52.5	63.5
200 ep + recovery stack (1 seed)	37.9	52.5	63.5
R-50 full method (reference)	**38.8**	**53.8**	**64.8**

*Official/offshore means of the 3-seed control axis; within the ±0.3 noise band of the 45.6/59.6 figures of Table I.

denoising entirely.

The consensus vote resolves this trade-off by gating on cross-view agreement: clusters with support in the band $2 \leq |C| \leq S_{\max}$ are majority-denoised, all others emit their seed mask verbatim. It matches or exceeds the seed vote on every split (38.8/53.8/64.8; inshore AP_{75} 36.1 against 35.3 for the seed vote) and behaves monotonically in the band ceiling: $S_{\max} = 4/5/6$ yields inshore 38.4/38.6/38.8 with offshore pinned at 64.8 throughout.

E. Architecture-Side Controlled Study

We frame this study as the regime experiment: on the same strong COCO-initialized base, we attempted seven architecture- and training-side directions under matched protocol, hunting for any modification worth its complexity. Table V and Fig. 9 record the outcome: all seven were flat, negative, or insufficient, which we report as controlled evidence for the regime diagnosis, under the same seed discipline as the positive results.

The mechanisms differ by direction. OffsetAdj constrains the sampling offsets of the multi-scale deformable attention and costs −1.0 inshore: the sampling geometry is exactly what COCO pretraining has already tuned, and constraining it perturbs the alignment. CHART, a one-to-many auxiliary supervision on the decoder, collapsed validation AP to 0.18 at auxiliary weight 1.0 and was flat at 0.1. CAST, a hard-sample paste augmentation, was flat at 27.7, absorbed by the strong baseline, as were all four training-side families (capacity, feature enrichment, data augmentation, supervision). A MaskIoU rescoring head is worth +1.5 inshore and +1.7 official standalone at its $\lambda = 0.2$ operating point, yet leaves a +0.1 residual under the stack: the weight soup subsumes its gain. The SGA strip adapter deserves emphasis because it preserves the COCO initialization exactly: sequential $1 \times 7/7 \times 1$ depthwise strip convolutions, consistent with the sequential-decomposition guidance of [8], sit behind a scalar zero-initialized residual gate [11], so the model starts fine-tuning as the identical COCO network. It still lost 0.9 inshore while gaining 0.6 offshore: the elongation prior fits large well-resolved offshore hulls and mismatches the quasi point-like inshore targets. Preservation alone does not rescue a mismatched prior.

The backbone swap is the most instructive arm. Swin-S carries a +3.2 COCO mask-AP advantage over ResNet-50 (46.1 against 42.9), which shrinks to +0.9 inshore on SAR (28.6 against 27.7), the only positive base-level delta in the sweep. Under the recovery stack (single seed, no soup) it reaches 37.3/52.5/63.5, below the ResNet-50 full method on all three splits. COCO backbone rankings do not transfer to this domain, and the base-level margin does not survive the pipeline.

The equal-epoch control asks whether the gap to 500-epoch competitors is a schedule artifact. We extended training to 200 epochs: the validation trajectory ran 43.5, 44.8, 44.9, saturating near epoch 45, and an early-decayed warm restart reached a best official base of 46.4, a +0.8 gain over the 55-epoch

TABLE VI
EFFECT OF THE OPTIONAL BOUNDARY-PATCH REFINER ON THE FULL STACK (MASK AP ×100)

Split	Configuration	AP	AP_{50}	AP_{75}	AP_s
inshore	full stack	38.8	75.6	36.1	35.8
inshore	+ refiner	**43.6**	76.3	**47.0**	**41.4**
official	full stack	53.8	89.4	62.2	52.8
official	+ refiner	**58.6**	89.9	**70.7**	**57.8**
offshore	full stack	64.8	98.5	81.7	64.1
offshore	+ refiner	**69.1**	98.6	**87.4**	**68.7**

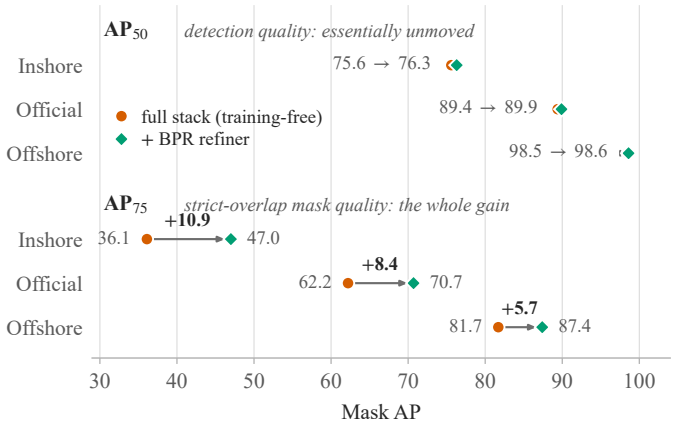

Fig. 10. Mask AP_{50} and AP_{75} before and after boundary-patch refinement (Table VI); each pair runs from the training-free full stack to the refined result. The strict-overlap gains of +10.9/+8.4/+5.7 on the inshore/official/offshore splits, set against AP_{50} pairs that barely separate, are the fingerprint of a pure boundary correction.

45.6 that concentrates in AP_{75} (51.5 against 49.4). With the recovery stack applied (single seed; a soup at 3×200 epochs is unaffordable), the 200-epoch model reaches 37.9/52.5/63.5, below the 55-epoch three-seed configuration at 38.8/53.8/64.8. Longer training does not close the inshore gap: the marginal base gain neither survives the stack nor compensates for the missing soup.

F. Boundary Refinement Analysis

The stack's residual error is boundary quality. Mask logits are predicted at stride 4 (200×200) and bilinearly upsampled $4\times$ to 800×800, so predicted contours carry a soft halo of roughly one to two pixels; on a typical 9×18-pixel inshore hull, a one-pixel boundary shift costs roughly 0.15–0.25 IoU. No training-free operator we tested moves this residual: binarization-threshold, probability-averaging, interpolation, and radiometric re-ranking controls were null or negative. We therefore added the optional trained tier of Section III-F: a boundary-patch refiner in the BPR paradigm [18], a 2.35M-parameter UNet trained once, in about 19 minutes, on (prediction, ground-truth) boundary patches harvested from the training split, and applied under a $\pm$4-pixel band guard that leaves every out-of-band pixel and every detection score untouched.

Table VI and Fig. 10 quantify the effect. The refiner adds +4.8/+4.7/+4.4 AP and +10.9/+8.4/+5.7 AP_{75} on inshore/official/offshore, respectively (deltas computed from unrounded values). The metric fingerprint is exactly that of a boundary correction: inshore AP_{50} moves by only +0.7 (75.6 to 76.3) while inshore AP_{75} moves by +10.9 (36.1 to 47.0). Detection and recall composition are untouched. The gain concentrates entirely at strict IoU, its magnitude matches the halo arithmetic above (a one-to-two-pixel correction is precisely the band in which instances cross the 0.75 threshold), and inshore AP_s rises from 35.8 to 41.4, confirming that the correction lands on the small near-shore hulls the diagnosis targeted.

A post-processor trained on (prediction, ground-truth) pairs invites the suspicion that test ground truth leaks into the output, so we audited the application path along seven attack vectors; all are clean. (1) No ground-truth access: the apply path receives no annotations, and the refined result regenerates MD5-identically from predictions, refiner weights, and images alone. (2) Band confinement: in 90 of 90 sampled instances, zero edited pixels fall outside the $\pm$4-pixel band. (3) Score integrity: 0 of the roughly 237 000 detection scores change. (4) Determinism: an independent re-run reproduces every reported AP to four decimals. (5) Visual evidence: refined contours track image scattering evidence, and false-positive masks are also edited toward that evidence, which would be impossible if the refiner copied ground truth, since false positives have none. (6) Split hygiene: the filename overlap between refiner training patches and test images is zero. (7) Metric fingerprint: the +0.7 AP_{50} versus +10.9 AP_{75} signature excludes any mechanism that adds, removes, or re-scores detections. One honesty note remains: the inshore row, on which the headline claim rests, received the full audit; the official and offshore rows were verified by the same band- and score-integrity spot checks rather than an exhaustive per-instance re-derivation.

G. Learned Re-Ranking Analysis

The stack and the refiner leave a third residual, and it is not in the masks. It is in the order the masks are ranked in. Mask2Former scores an instance as the product of its classification confidence and the mean foreground confidence of its mask, a quantity that never compares that mask with the object it covers, so a tight hull and a loose one at equal class confidence are ranked alike; average precision is an integral over a ranking, so a re-ordering alone moves it. Section III-G specifies the operator that exploits this. A gradient-boosted regression tree ensemble over the 14 scalars of Eq. (12), six of which the consensus fusion has already computed, is fitted on the pipeline's own training-split candidates against their mask IoU with the annotation and blended into the score by Eq. (14), at a weight chosen on 728 held-out training images. It reads no image pixel and it fits in 19 seconds on CPU. This subsection reports what it buys, what it costs on the columns it does not help, and whether it can be cheating.

Table VII reports the effect on top of the refined stack. Re-ordering alone is worth +1.0/+0.8/+1.0 AP inshore/official/offshore (+0.97/+0.82/+0.95 before rounding), +1.3/+0.7/+0.6 AP_{75}, and +0.6/+0.6/+0.7 AP_s, at +0.4/+0.1/+0.1 AP_{50}. Nothing is added, removed, or reshaped:

TABLE VII
EFFECT OF THE OPTIONAL LEARNED RE-RANKER ON THE REFINED STACK (MASK AP $\times 100$). ONLY SCORES ARE EDITED; EVERY MASK IS BIT-IDENTICAL TO THE REFINER ROW ABOVE IT

Split	Configuration	AP	AP_{50}	AP_{75}	AP_s
inshore	+ refiner	43.6	76.3	47.0	41.4
inshore	+ re-ranking	**44.5**	**76.7**	**48.3**	**42.0**
official	+ refiner	58.6	89.9	70.7	57.8
official	+ re-ranking	**59.4**	**90.1**	**71.4**	**58.3**
offshore	+ refiner	69.1	98.6	87.4	68.7
offshore	+ re-ranking	**70.1**	98.6	**88.0**	**69.4**

Bold denotes the better value of each pair. The offshore AP_{50} pair reads 98.56 against 98.63 before rounding and is left unmarked.

the same detections match the same annotations at the same IoU levels, and only their position in the score-descending list changes, which is why the movement is small, positive at every threshold, and evenly spread rather than concentrated at strict overlap the way the boundary correction was. One column moves the other way and we report it plainly. Large-object AP falls by 1.4, 1.3, and 1.3 points on the three splits (48.3 to 47.0 inshore). The regressor is fitted on all training candidates, which are overwhelmingly small vessels, and it demotes a few of the rare large instances in exchange for promoting many small ones; on HRSID that trade is strongly net positive, and on a large-vessel-dominated corpus it would need to be re-examined.

A component trained on prediction and ground-truth pairs and then applied to test predictions invites the same suspicion the refiner did, so we audited it along six vectors; all are clean. (1) No leakage: the regressor is fitted only on training-split predictions and training-split annotations, the apply path receives no test annotation of any kind, and every feature it consumes is computable from the prediction alone. (2) Masks untouched: not one mask byte changes, the run-length encodings and the cardinality of the candidate set are identical to the refined input, and only the score field is rewritten; detections that find no match in the stack dump keep their original score, which is 4682 of 28 539 inshore, 55 348 of 118 922 official, and 50 666 of 90 383 offshore. (3) Determinism: an independent re-run regenerates the output files byte-identically and reproduces every reported AP. (4) Predictive validity: on held-out training images the predicted mask IoU tracks the realized one at a Spearman rank correlation of 0.68 (Section III-G), so the component ranks by mask quality rather than by a memorized identity. (5) One model, no per-split tuning: a single regressor with a single α is applied unchanged to all three splits, and α itself was chosen on held-out training images, never on test. (6) Resampling: across 1000 bootstrap resamples of the inshore test images the gain is positive in 100% of them, with a 95% confidence interval of [+0.61, +1.23].

Two variants on the same axis returned nothing, and both are informative. The first is a miss-rescue verifier: a small convolutional network re-scores the rejected tail of the candidate pool, those detections scoring in $[0.01, 0.35)$, and appends whatever it endorses at a capped score below the kept floor. It rescued 9402 inshore, 10 145 official, and 743 offshore detections and moved AP to 43.5/58.5/69.1 against 43.6/58.6/69.1, a null within three hundredths of a point and marginally negative; stacked on the re-ranker it repeats the null. The mechanism is that our dumps already run down to score zero at 100 detections per image, so a ship the pipeline has seen at all is already somewhere in the ranked list. There is no missing detection to rescue, only a badly placed one to promote, which is what the re-ranker does. The second variant extends the feature vector from 14 to 16 with two depth-consistency descriptors, the mean mask IoU of the seed instance across the last three decoder layers and its score trajectory across them, on the hypothesis that an instance stable across decoder depth deserves trust. It ties: 44.5/59.3/70.0 against 44.5/59.4/70.1. Stability across decoder depth and agreement across views carry the same information on this data, because the late decoder layers converge on exactly the instances the views already agree about, so the two extra columns are redundant with the six consensus features that are there. We report both nulls because the axis they failed on is the axis the re-ranker succeeded on, and the contrast is the evidence that ranking, not recall, was the residual.

H. Qualitative Analysis

Fig. 11 traces where the recovered AP lives at instance level, on four test scenes ordered by target density. The dense harbor of row (a) holds 88 annotated ships; the base finds 31 of them while raising 37 false alarms and missing 57. The training-free stack lifts the same scene to 72 true positives, 28 false alarms, and 16 misses: 41 of the 57 missed targets are recovered while false alarms drop from 37 to 28. Movement in both directions at once is the fingerprint of view-consensus filtering; a lowered decision threshold would buy recall with additional false alarms, whereas mask-level voting across the six views admits the dim, consistently re-detected hulls and suppresses the view-unstable clutter responses. The quay line of row (b) repeats the pattern at smaller scale, from 17/19/7 (TP/FP/FN) at the base to 23/12/1 under the stack, and the refiner clears the final miss to 24 of 24; the parallel-berthed group of row (c) goes from 5 of 7 to 7 of 7. In the offshore scene of row (d), all three configurations detect all 5 ships with no false alarms, consistent with the near-saturated offshore AP_{50} of Table VI; what remains there is boundary quality, and the inset magnifies the true positive whose mask the refiner edits most, trimming a loose stern boundary (the stack mask and its refined version agree at only 0.685 IoU on that hull).

The failure modes are equally visible. Residual red contours in rows (a)–(c) sit almost exclusively on bright quay structures and unlabeled lookalike clutter, the same radiometrically ambiguous responses that survive consensus because they persist across views; at the 0.3 display threshold the stack still reports 28, 12, and 5 false alarms in the three inshore scenes (one of them, in row (c), a duplicate fragment inside the largest ship that persists across the prediction columns at this display threshold). The refiner itself changes counts only at the margin, as its band guard promises: in row (b) one borderline hull crosses the 0.5 matching threshold upward (23 to 24 true positives), while in row (a) one crosses it downward (72 to

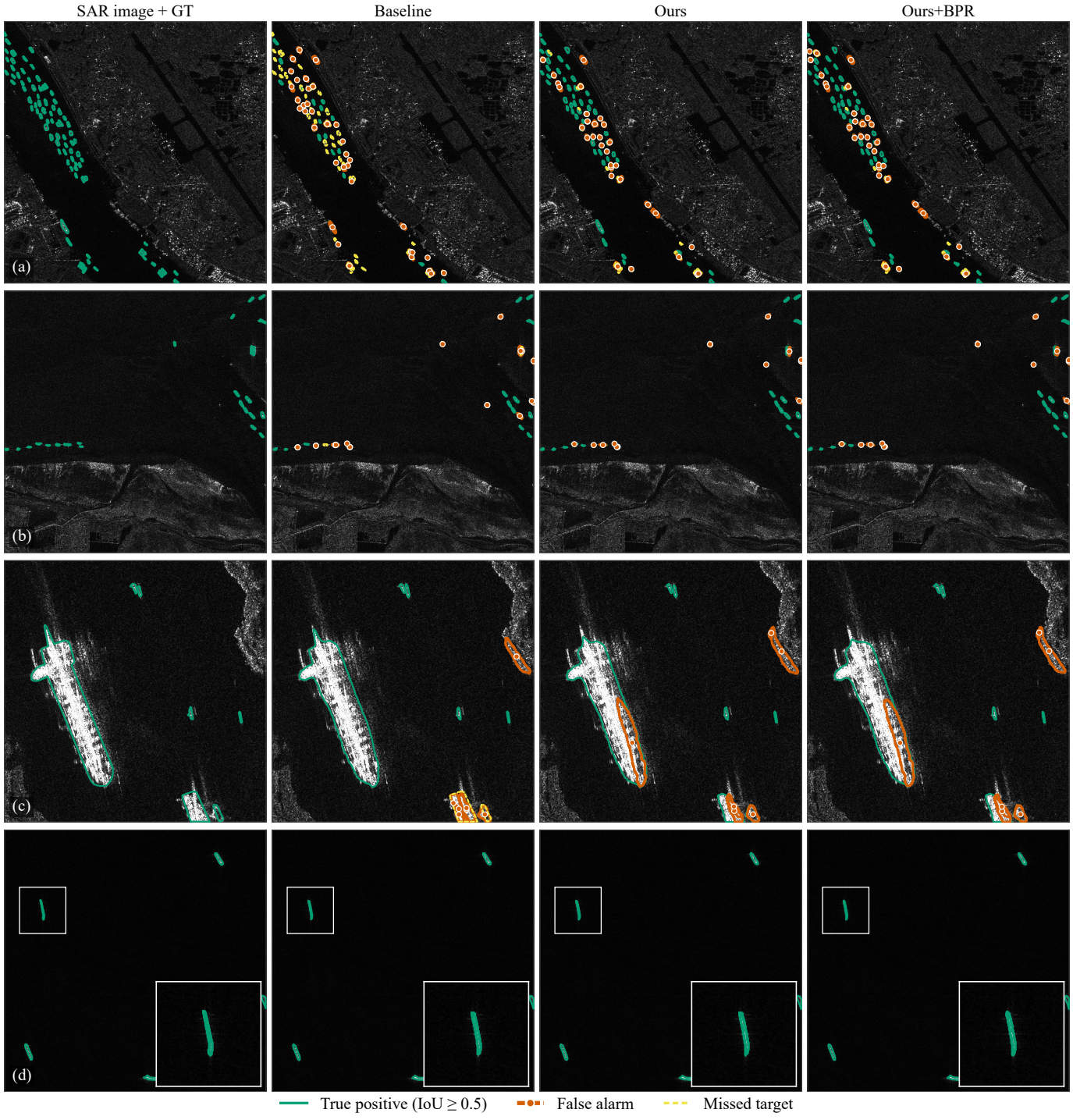

Fig. 11. Qualitative comparison on four HRSID test scenes at a 0.3 score display threshold. Columns show, from left to right, the SAR image with ground truth, the base, the training-free stack, and the stack with the boundary-patch refiner. Green contours are true positives (mask IoU ≥ 0.5 against ground truth), red contours false alarms, and yellow dashed contours missed targets. Rows: (a) dense harbor; (b) quay line; (c) parallel-berthed group; (d) offshore scene, with the inset on the true positive whose mask the refiner edits most.

71), a reminder that a pure boundary edit can move an instance across a fixed matching threshold in either direction while the AP gains of Table VI accrue at strict IoU.

I. Comparison With Independently Trained Detectors

Section IV-C compares against published HRSID figures produced under a different protocol, and Section IV-H compares the pipeline against its own base. Neither puts the pipeline and a competing detector on the same scene at the same budget, so Fig. 12

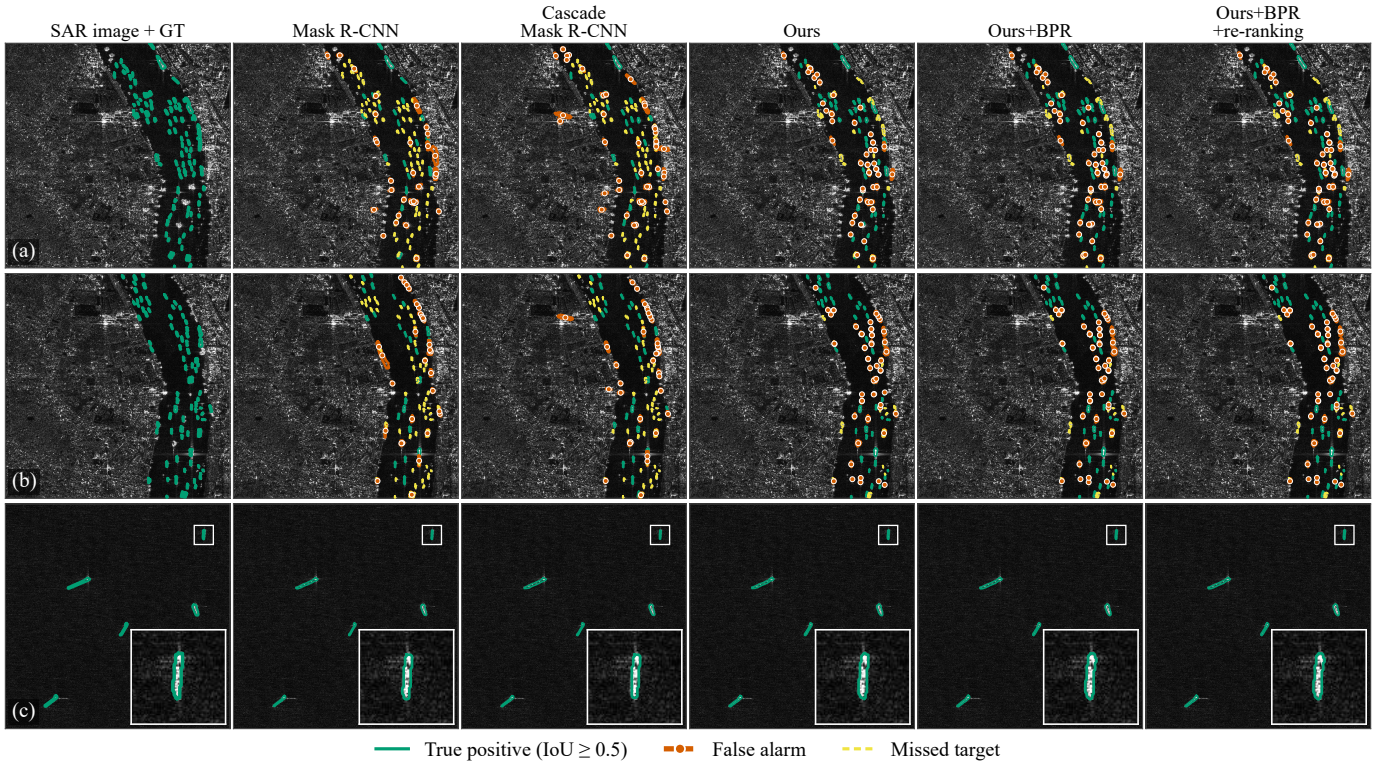

Fig. 12. Six-column qualitative comparison on three HRSID test scenes at a 0.3 score display threshold and mask IoU ≥ 0.5 matching; color coding as in Fig. 11. Columns show, from left to right, the SAR image with ground truth, Mask R-CNN, Cascade Mask R-CNN, the training-free stack, the stack with the boundary-patch refiner, and the same with the learned re-ranker. Rows (a) and (b) are inshore harbor scenes; row (c) is offshore, where the $4.9\times$ inset magnifies the hull on which the methods disagree most.

does that on three HRSID test scenes: inshore chips `P0115_4800_5600_9600_10400` with 95 annotated ships and `P0111_5400_6200_9600_10400` with 88, and offshore chip `P0072_1200_2000_5400_6200` with 5. The 88-ship chip of row (b) is a different scene from the 88-ship harbor of Fig. 11(a). Scene selection is programmatic and reproducible: the two inshore scenes maximize one point per recovered target plus half a point per removed false alarm against the better of the two competitors on that same scene, subject to at least 8 annotated ships and at most five false alarms above that competitor, and the offshore scene is, among the 20 scenes that clear a filter requiring at least five annotated ships and zero misses and zero false alarms from every method, the one with the largest cross-method spread in mean mask IoU. The Mask R-CNN and Cascade Mask R-CNN columns are our own trainings on the identical 3642-image training split, for the identical 55 epochs, scored by the identical evaluator; they are two of the three HRSID carriers of Table X, whose carrier-alone inshore AP reads 28.3 and 30.4. A third detector trained under that same protocol, SOLOv2, is the weakest of the three and is omitted from the figure for width: it reads 19.0 inshore, 38.2 official, and 54.0 offshore mask AP, with an inshore AP_{75} of 15.7 against 24.8 and 27.0 for the two columns shown, and it is carried through the same three-rung ladder in Table X. On the 95-ship chip of row (a) the two competitors find 23 and 28 of the annotated ships while raising 32 and 44 false alarms, and the training-free stack finds 66 at 34. On the 88-ship chip of row (b) they find 25 and 34 at 29 and 33 false alarms, and the stack finds 65 at 35. The complete per-panel accounting, in the column order Mask R-CNN / Cascade Mask R-CNN / Ours / Ours+BPR / Ours+BPR+re-ranking, reads 23/32/72, 28/44/67, 66/34/29, 65/35/30, and 65/35/30 true positives, false alarms, and missed targets on row (a) and 25/29/63, 34/33/54, 65/35/23, 65/35/23, and 65/35/23 on row (b), with mean matched mask IoU of 0.692/0.703/0.704/0.768/0.770 and 0.696/0.703/0.734/0.788/0.786, respectively. The false-alarm column barely moves across that transition, 32 and 44 against 34 and 35, so between 1.9 and 2.9 times as many targets are found at a comparable false-alarm count. That arithmetic is the signature of recall recovery rather than of a lowered decision threshold, which would have raised both columns together, and it is the same recall-and-precision-together behavior that Section IV-H reported against our own base, now measured against detectors trained under our protocol rather than against an earlier rung of our own pipeline.

The two trained tiers then act where the counts have stopped moving. In row (a) the refiner and the re-ranker hold 65 true positives against the stack's 66, one instance crossing the 0.5 matching threshold downward, while the mean mask IoU of the matched instances rises from 0.704 to 0.768 and then 0.770; in row (b) all three of our columns read 65/35/23 and the mean matched IoU rises from 0.734 to 0.788, with the re-ranker at 0.786. Flat counts with rising overlap are

the instance-level form of the AP_{50}-flat, AP_{75}-up pattern that Tables VI and VII report. Row (c) removes detection from the comparison altogether: all five method columns find all 5 offshore ships with no false alarms, so boundary quality is the only quantity left, and it separates them at 0.782 and 0.777 for the two competitors against 0.828 for the stack and 0.873 for both trained tiers, a cross-method spread of 0.263 on the hull the inset magnifies. One caveat travels with the figure. The competitor columns are trained at our budget and not at theirs, so what is shown is a same-protocol comparison and not a statement about the published systems of Table II, whose 500-epoch scratch rows score higher in absolute terms than the carriers plotted here.

J. Cross-Dataset Generalization

The recipe transfers to PSeg-SSDD with one proportional change. We retrained only the base under the protocol of Section IV-A: three seeds (s0/s2/s3) of the identical COCO-initialized Mask2Former, 8352 iterations each at an effective batch of 4, that is, 36 epochs over the 928-image official training split, averaged into a weight soup; the 232-image test set is partitioned into 46 inshore and 186 offshore scenes following [2]. The wide-TTA scale set is rescaled in proportion to the smaller native chips, from $\{640, 800, 960, 1088\}$ to $\{416, 512, 608, 704\}$; the consensus band $2 \leq |C| \leq S_{\max} = 6$ is carried over unchanged; the boundary-patch refiner is retrained for 20 epochs on patches harvested from SSDD training-split predictions, with the ± 4-pixel band guard unchanged; and the re-ranker is refitted on SSDD training-split predictions from the identical 14 features, with its blend weight re-selected by the same held-out sweep, which returns $\alpha = 1.0$ here, so the SSDD re-ranker replaces the score outright instead of blending it. Table VIII reports the outcome. The soup base scores 60.9/51.9/64.8 mask AP on the official/inshore/offshore splits; the training-free stack lifts it to 63.3/54.0/66.9, the refiner to 66.3/60.0/68.8, and the re-ranker to 67.5/62.1/69.8, a composed lift of +6.6/+10.2/+5.0 (stack +2.4/+2.1/+2.1, refiner +3.0/+6.1/+1.8, re-ranking +1.2/+2.1/+1.1; deltas computed from unrounded values). Every operator keeps the sign it had on HRSID. The refiner again concentrates at strict overlap, raising inshore AP_{75} from 54.9 to 72.5 and official-split AP_{75} to 82.9, and the re-ranker carries those two figures to 74.6 and 84.9. A fusion-ordering probe also replicates: majority pixel voting reads 56.9 on the SSDD inshore split against 54.0 for both the consensus and seed votes, the same inshore lead it shows on HRSID, and we retain the consensus operator because the offshore-erosion cost of unconditional voting was established there (Section IV-D).

The transfer also fixes the position of the method against specialized architectures on this dataset. Under the same split protocol, the complete pipeline exceeds the from-scratch Mask2Former baseline tabulated by O2Former (52.5 inshore, 67.2 offshore) by +9.6/+2.6, and closes to 1.1 inshore and 2.1 offshore points below the 500-epoch specialized architecture itself (63.2/71.9) [2]; the near-shore gap that stood at 3.2 points after the refiner is 1.1 after re-ranking. The magnitude of each tier tracks regime difficulty. SSDD single-view predictions are more stable than their HRSID counterparts (base inshore 51.9 against 27.7), so the variance-driven stack has less to remove: its inshore lift is +2.1 here against +11.1 on HRSID. The boundary tier moves in the opposite direction and dominates the SSDD composition, contributing +6.1 of the +10.2 inshore lift and +17.6 inshore AP_{75}. Fig. 13 traces the same behavior at instance level, now against the competing architectures rather than against our own tiers. In the inshore harbor of row (a), Mask R-CNN, CondInst and M2F+OAEM find 7, 5 and 6 of the 10 annotated ships while the full method finds all 10; the recall is bought with false alarms, 12 against the 7 of M2F+OAEM, the recall-for-consensus trade of Section IV-H at smaller chip scale. Row (b) is the sharper case: on 13 ships along a quay the two conventional detectors collapse to 4 and 3 true positives, M2F+OAEM reaches 11 with 5 false alarms, and the full method reaches 13 with the same 5 false alarms, so the recall gain there is free. Row (c) isolates the boundary tier: all five methods detect all 7 offshore ships with no false alarms, and only mask quality separates them, with mean IoU against ground truth of 0.888, 0.836, 0.859, 0.867 and 0.898 in column order. The refiner therefore also wins the scene in which detection is already saturated, and the inset magnifies the hull on which the five columns disagree most. The complete per-panel accounting, in the column order Mask R-CNN / CondInst / M2F+OAEM / Ours / Ours+BPR, reads 7/7/3, 5/0/5, 6/7/4, 10/12/0, and 10/12/0 true positives, false alarms, and missed targets on row (a), 4/4/9, 3/3/10, 11/5/2, 13/5/0, and 13/5/0 on row (b), and 7/0/0 for every method on row (c); the M2F+OAEM column is our own reimplementation of that module on our carrier and not the published O2Former system. Scene selection is again programmatic: the two inshore scenes maximize the missed-target gap against the strongest competitor, M2F+OAEM, subject to at least 8 annotated ships, and the offshore scene is the one detected in full by every method with the highest Ours+BPR mask IoU.

Because PSeg-SSDD is small enough to retrain competitors on our own hardware, we complement the relayed literature numbers of the preceding paragraph with a comparison in which every row is ours. Table IX reports six independently trained methods spanning four families: a two-stage region-based detector and its deformable-convolution variant, a single-stage prototype detector, a single-stage dynamic-kernel detector, and two SAR-specific modules grafted onto the same Mask2Former carrier that our own base uses. Neither of those

TABLE VIII
TRANSFER OF THE COMPLETE RECIPE TO PSEG-SSDD (MASK AP $\times 100$). ONLY THE TTA SCALES ARE RESCALED TO THE NATIVE CHIP SIZE; "–": FULL METRICS NOT LOGGED

Configuration	Official	Inshore	Offshore	In. AP_{75}
soup base (avg. s0/s2/s3, no TTA)	60.9	51.9	64.8	–
+ wide TTA + consensus vote	63.3	54.0	66.9	54.9
+ BPR refiner (trained tier)	66.3	60.0	68.8	72.5
+ learned re-ranking (trained tier)	**67.5**	**62.1**	**69.8**	**74.6**

Bold denotes the best value in each column. Both trained tiers are refitted on SSDD training-split predictions; no operator is retuned.

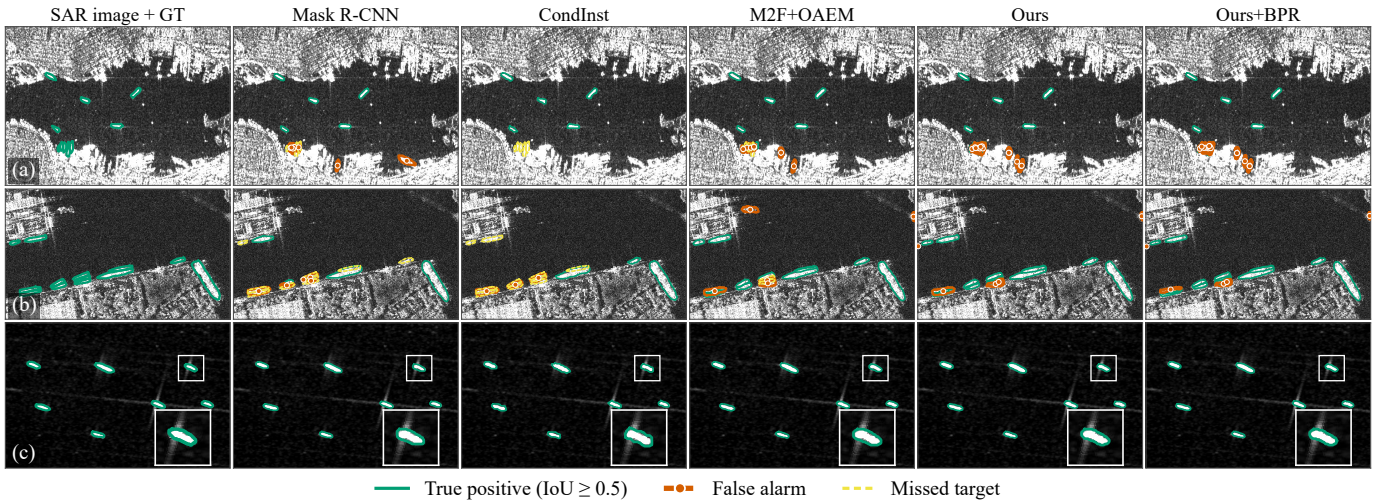

Fig. 13. Six-column qualitative comparison on three PSeg-SSDD test scenes at a 0.3 score display threshold and mask IoU ≥ 0.5 matching; color coding as in Fig. 11. Columns show, from left to right, the SAR image with ground truth, Mask R-CNN, CondInst, M2F+OAEM, the training-free stack, and the stack with the boundary-patch refiner. Rows: (a) inshore harbor; (b) inshore quay; (c) offshore, where every method detects every ship and the inset magnifies the hull with the largest inter-method mask-IoU spread.

TABLE IX
COMPARISON WITH SIX INDEPENDENTLY TRAINED METHODS ON PSEG-SSDD (MASK AP $\times 100$). EVERY ROW, COMPETITORS INCLUDED, WAS TRAINED AND SCORED BY US UNDER ONE EVALUATOR. ROWS ARE ORDERED BY INSHORE AP

Method	Family	Init	Epochs	Inshore	Official	Offshore
YOLACT [21]	single-stage prototype	ImageNet	55	38.9	55.7	63.4
Mask R-CNN [22]	two-stage region-based	ImageNet	50	40.8	57.8	65.1
Mask R-CNN + DCN [23]	two-stage + deformable conv. (DCN)	ImageNet	50	41.0	57.8	64.6
CondInst [24]	single-stage dynamic kernel	COCO	36	41.2	54.3	60.1
M2F + DenoDet-style NDM (our reimpl.)* [25]	query-based + SAR module	COCO	36	52.9	60.5	63.7
M2F + OAEM (our reimpl.)* [2]	query-based + SAR module	COCO	36	53.2	60.8	64.0
Ours, soup base	query-based, unmodified	COCO	3×36	51.9	60.9	64.8
Ours, full method	+ training-free stack	COCO	3×36	54.0	63.3	66.9
Ours, full method + BPR	+ boundary refiner	COCO	3×36 (+20 BPR)	60.0	66.3	68.8
Ours, full method + BPR + re-ranking	+ learned re-ranking	COCO	3×36 (+20 BPR)	**62.1**	**67.5**	**69.8**

Bold denotes the best value in each column. All rows were fine-tuned on the identical 928-image training split and scored with the identical COCO evaluator at 512×512; the two budget and initialization groups are stated in the Init and Epochs columns, and the ImageNet group received the larger epoch budget. *No public code exists for either module, so these two rows are our own reimplementation of the published descriptions on our Mask2Former carrier, applied as a zero-initialized gated residual so that step 0 reproduces the carrier exactly. They measure that reimplementation, not the published systems. Three further methods were run and excluded: SOLO diverged, its training loss rising from 3.68 to 10.68 with zero detections at all 50 validation epochs; PointRend and InstaBoost were trained at 1333×800 but tested at 512×512, so their scores are a resolution artifact rather than evidence about the mechanism; FreqFusion collapsed to 3.6 AP in its own training log.

two modules has public code, so both were reimplemented by us from their published descriptions; the rows report our reimplementation on our carrier and not the published systems, and they are labeled accordingly in the table. The training-free full method exceeds all six on all three splits, and with both trained tiers the inshore margin over the strongest of them, our OAEM reimplementation at 53.2, is +8.9, while the official and offshore margins are +6.7 over M2F + OAEM (60.8) and +4.7 over Mask R-CNN (65.1). The ranking inside the competitor block is itself informative. The four generic architectures cluster between 38.9 and 41.2 inshore, a spread of 2.3 points across two-stage, prototype, and dynamic-kernel designs, whereas the two SAR-specific rows jump to 52.9 and 53.2, so on this dataset the near-shore regime, not the choice of segmentation head, sets the level. Our own unmodified base already reaches 51.9 inshore with no SAR-specific component at all, within 1.3 points of the better of the two specialized rows, and the training-free stack passes both without touching the model.

The two SAR-specific rows carry the most weight in Table IX, because they change the architecture exactly where we do not. As published, OAEM enriches the multi-scale backbone features with an oriented aggregation along the hull axis and a polar position encoding [2], and the frequency-domain module of DenoDet V2 soft-thresholds the amplitude spectrum to suppress multiplicative speckle while preserving phase [25]. What Table IX measures is our own reimplementation of those two descriptions, written because no code was released for either, and any shortfall against the authors' intent is ours rather than theirs. Both reimplementations sit on

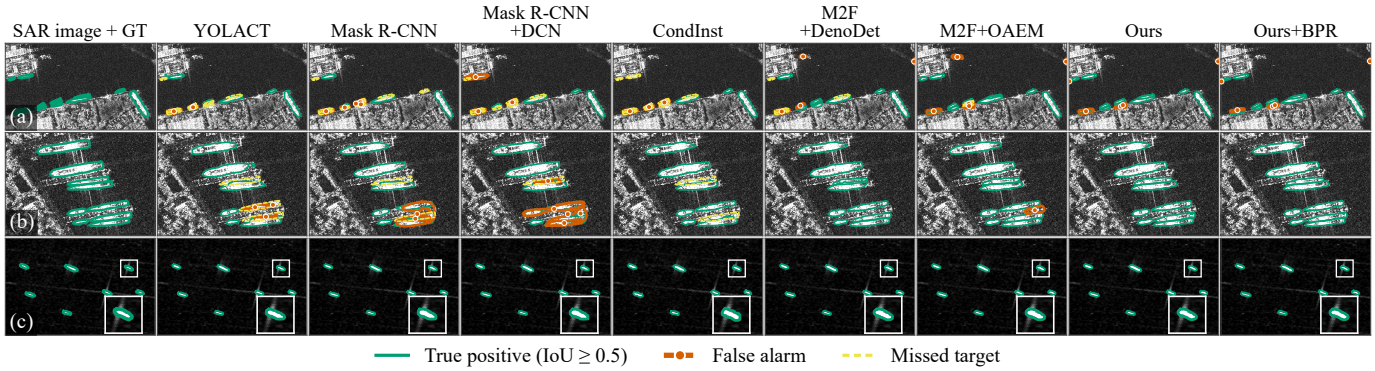

Fig. 14. Nine-column qualitative comparison on three PSeg-SSDD test scenes at a 0.3 score display threshold and mask IoU ≥ 0.5 matching; color coding as in Fig. 11. Columns show, from left to right, the SAR image with ground truth, YOLACT, Mask R-CNN, Mask R-CNN+DCN, CondInst, M2F+DenoDet, M2F+OAEM, Ours, and Ours+BPR. Rows (a) and (c) repeat Fig. 13(b) and (c) with eight method columns instead of five; row (b) is a further inshore scene.

TABLE X
UNIVERSALITY OF THE INFERENCE-SIDE RECIPE ACROSS FROZEN CARRIERS (MASK AP $\times 100$). EACH CARRIER IS TRAINED ONCE AND THEN FROZEN; "+STACK" APPLIES THE WIDE EIGHT-VIEW MULTI-SCALE FLIP TTA WITH CONSENSUS-VOTE FUSION, "+BPR" ADDITIONALLY APPLIES A PER-CARRIER BOUNDARY-PATCH REFINER, AND "+R11" ADDITIONALLY APPLIES THE LEARNED RE-RANKER

Carrier	Variant	Inshore	Full test	Offshore
HRSID (full test = official test set)				
Mask R-CNN	carrier alone	28.3	44.8	57.8
	+stack	37.1	52.2	63.6
	+BPR	40.2	56.2	68.2
Cascade R-CNN	carrier alone	30.4	46.5	59.2
	+stack	39.1	53.3	64.0
	+BPR	42.3	57.6	69.0
SOLOv2	carrier alone	19.0	38.2	54.0
	+stack	25.5	43.0	56.3
	+BPR	33.3	53.0	67.6
M2F (ours)	carrier alone	29.7	47.6	61.4
	+stack	38.8	53.8	64.8
	+BPR	43.6	58.6	69.1
	+R11	44.5	59.4	70.1
PSeg-SSDD				
Mask R-CNN	carrier alone	40.8	57.8	65.1
	+stack	44.1	58.6	64.5
	+BPR	47.7	61.3	66.5
CondInst	carrier alone	41.2	54.3	60.1
	+stack	50.1	59.7	63.6
	+BPR	57.5	66.1	69.3
YOLACT	carrier alone	38.9	55.7	63.4
	+stack	41.1	56.2	62.9
	+BPR	45.0	59.7	66.1

Every row is scored with the identical COCO evaluator against the identical ground truth, and the recipe is applied with no per-carrier tuning of any kind. The M2F carrier-alone row is the weight soup with the λ=0.2 mask-IoU rescoring that the stack configuration carries (Section IV-E), hence 29.7/47.6/61.4 rather than the 29.6/47.4/61.1 of Table I.

the identical COCO-initialized Mask2Former carrier as our base and were trained for the same 36 epochs on the same split. Measured against that shared carrier, they read +1.3 and +1.0 inshore but -0.1 and -0.4 official and -0.8 and -1.1 offshore, so each buys a near-shore gain at the price of the other two splits. The training-free stack reads +2.1/+2.4/+2.1 on the same base and keeps the sign on all three. Architectural specialization is therefore not the only way to reach the near-shore regime on this dataset, and in this comparison it is not the more economical one. One selection detail is disclosed rather than corrected: the OAEM and DenoDet checkpoints were selected on the inshore split, which is the split their rows lead, whereas ours follow the best-segmentation-AP convention on the reported split. Both conventions select on the split they report, so the practice is symmetric across rows, but the two specialized inshore entries should still be read as upper bounds.

Fig. 14 carries the same eight methods to instance level, and the honest reading of it is narrower than the margins of Table IX. Its three scenes are `001091.jpg` with 13 annotated ships, `000219.jpg` with 9, and offshore `001131.jpg` with 7, the sole offshore scene clearing the same all-methods-

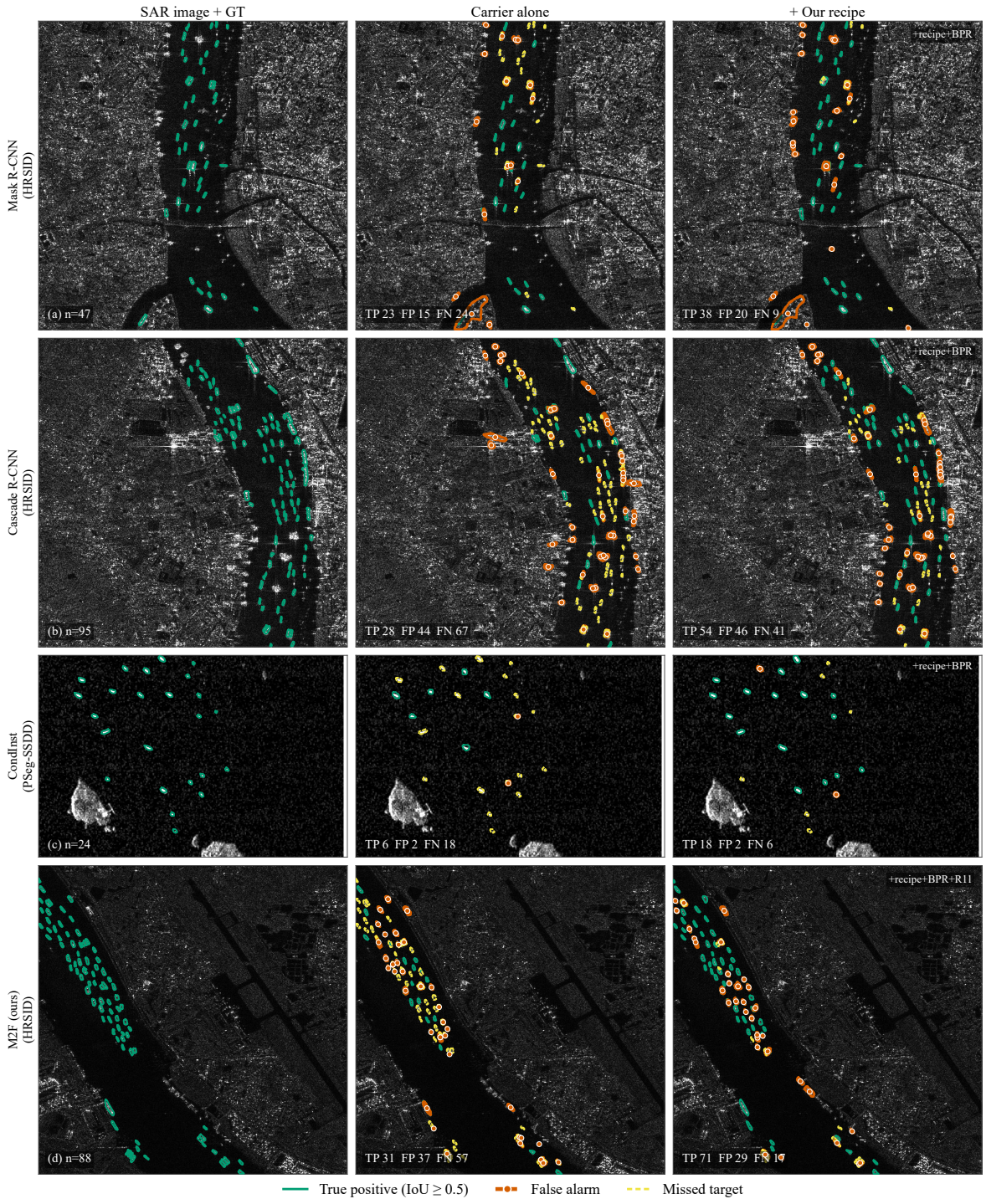

Fig. 15. The identical inference-side recipe applied to four frozen carriers, one per row, on the inshore split (Table X). Columns are the SAR image with ground truth, the carrier alone, and the same carrier with the recipe applied, at a 0.3 score display threshold; color coding as in Fig. 11. Rows: (a) Mask R-CNN on HRSID; (b) Cascade R-CNN on HRSID; (c) CondInst on PSeg-SSDD; (d) our own Mask2Former carrier on HRSID, the only row that additionally applies the learned re-ranker; rows (a)–(c) apply the boundary-refined stack.

complete filter used in Fig. 12. The per-panel true-positive, false-alarm, and missed-target counts, in the column order of the figure, read 5/3/8, 4/4/9, 5/3/8, 3/3/10, 9/4/4, 11/5/2, 13/5/0, and 13/5/0 on row (a) and 6/1/3, 7/2/2, 8/4/1, 7/0/2, 9/0/0, 9/1/0, 9/0/0, and 9/0/0 on row (b), while row (c) reads 7/0/0 for all eight methods and separates only on the mean mask IoU of the matched instances, at 0.877, 0.888, 0.878, 0.836, 0.863, 0.859, 0.867, and 0.898. Exactly one of the 46 inshore test scenes yields a strictly positive count gain for our columns over the best of the six competitors on that scene, the 13-ship quay of row (a), where the DenoDet and OAEM reimplementations reach 9 and 11 of the 13 ships and both

TABLE XI
FALSE-ALARM / MISSED-DETECTION ACCOUNTING ON PSEG-SSDD (MASK-IOU ≥ 0.5, GREEDY SCORE-DESCENDING MATCHING). LEFT BLOCK FIXES THE SCORE THRESHOLD AT 0.30; RIGHT BLOCK USES THE PER-METHOD, PER-SPLIT F_1-OPTIMAL THRESHOLD SWEPT OVER $0.05:0.05:0.90$

	Fixed threshold 0.30			F_1-optimal threshold			
Method	P_d	FA/img	F_1	P_d^*	FA/img*	F_1^*	thr*
Inshore split (46 images, 172 GT ships)							
YOLACT	0.709	0.957	0.722	0.692	0.587	0.748	0.40
Mask R-CNN	0.721	1.326	0.695	0.663	0.587	0.728	0.70
Mask R-CNN + DCN	0.738	1.739	0.670	0.663	0.717	0.715	0.65
CondInst	0.657	**0.217**	0.766	0.744	0.565	0.785	0.20
M2F + DenoDet-style NDM (our reimpl.)	0.924	1.000	**0.844**	0.843	**0.261**	0.881	0.85
M2F + OAEM (our reimpl.)	0.907	1.109	0.823	0.866	0.370	0.882	0.80
Ours, base	0.866	0.717	0.842	0.855	0.326	0.880	0.40
Ours, full method	0.977	1.435	0.828	0.924	0.565	0.891	0.55
Ours, full method + BPR	**0.983**	1.413	0.833	**0.936**	0.522	**0.902**	0.55
Offshore split (186 images, 374 GT ships)							
YOLACT	0.960	0.075	0.961	**0.992**	0.118	0.967	0.15
Mask R-CNN	0.933	0.081	0.946	0.928	0.048	0.951	0.75
Mask R-CNN + DCN	0.933	0.091	0.943	0.922	**0.038**	0.950	0.75
CondInst	0.928	**0.038**	0.953	0.976	0.081	0.968	0.10
M2F + DenoDet-style NDM (our reimpl.)	**0.987**	0.075	0.975	0.984	**0.038**	**0.983**	0.80
M2F + OAEM (our reimpl.)	**0.987**	0.054	**0.980**	0.984	**0.038**	**0.983**	0.85
Ours, base	0.963	0.167	0.941	0.936	0.075	0.949	0.40
Ours, full method	**0.987**	0.086	0.972	0.979	**0.038**	0.980	0.50
Ours, full method + BPR	**0.987**	0.086	0.972	0.979	**0.038**	0.980	0.50

$P_d = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$; the missed-detection rate is $P_m = 1 - P_d$ and is therefore omitted. $F_1 = 2\mathrm{TP}/(2\mathrm{TP}+\mathrm{FP}+\mathrm{FN})$.
Bold marks the best value in each column (P_d and F_1: higher is better; FA/img: lower is better).
Our own method names are set in bold face; all rows are our own runs under one identical protocol. No public code exists for the OAEM or DenoDet modules, so those two rows measure our reimplementation on our carrier rather than the published systems.
P_d alone is not comparable across rows at a fixed threshold, since it can be inflated by false alarms; read it together with FA/img, or use the F_1 columns.

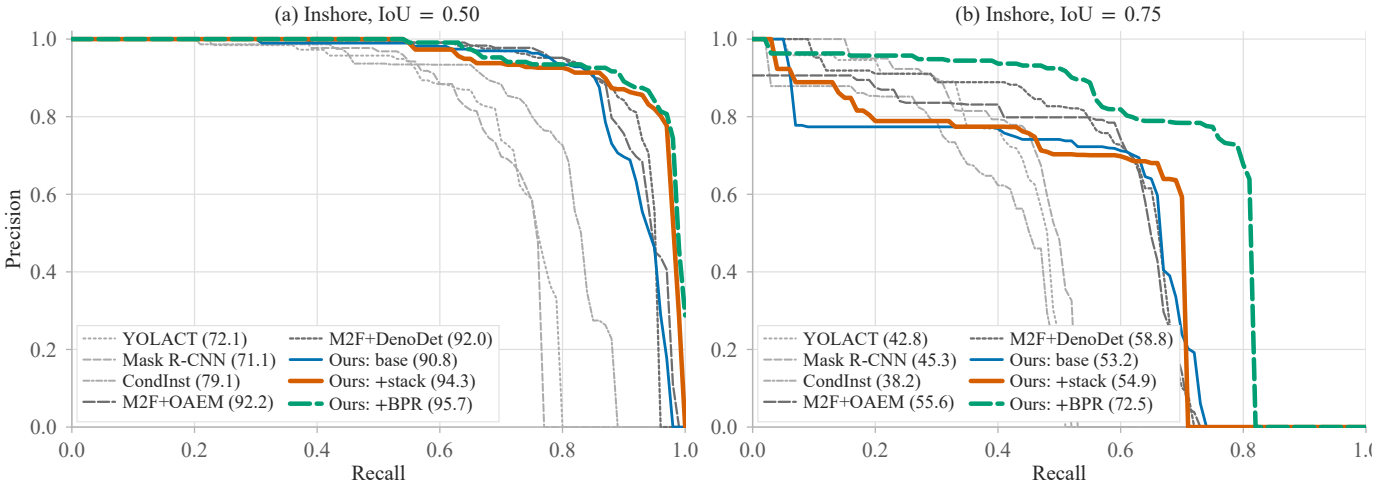

Fig. 16. Precision-recall curves on the PSeg-SSDD inshore split, recomputed with the COCO evaluator from the same prediction dumps as Tables IX and XI. (a) IoU $= 0.50$, which measures whether the ship is found. (b) IoU $= 0.75$, which measures whether its mask is accurate. The legend reports the area under each plotted curve, that is, the mask AP at the panel's IoU threshold. The inshore split is shown because offshore AP_{50} spans only 95.6 to 98.6 across the eight methods and the curves there collapse onto one corner.

of our columns reach 13 at the same five false alarms that OAEM already pays. Row (b) is a tie: the two SAR-module columns already find all 9 ships at the 0.3 display threshold, at zero and one false alarm, so our columns can separate only on overlap, at 0.823 and 0.844 mean matched IoU against 0.787 and 0.834. Row (c) is a tie for all eight methods at 7 of 7, with the refined column leading on mean matched IoU at 0.898 against a competitor best of 0.888. We state this rather than select around it, because it is the visible form of what Table IX already reports: on the easier dataset the specialized baselines sit close to detection saturation, and the margin the recipe carries there lives at the mask level and in the ranking, which is where its +6.1 inshore AP and +17.6 inshore AP_{75} from the boundary tier live as well.

K. Universality Across Architectures

Everything reported so far rides on one carrier, so the recipe could in principle be an artifact of Mask2Former. Table X settles that question on six further carriers spanning five architecture families across both datasets: a two-stage region-based detector, its cascade variant, a single-stage dynamic-kernel detector, a single-stage prototype detector, and a grid-based mask-kernel segmenter. Each carrier is trained once, frozen, and then handed the identical recipe with no per-carrier tuning of any kind, that is, the same eight views, the same consensus band, the same ± 4-pixel band guard, and a boundary refiner fitted by the same script on that carrier's own training-split errors. All 66 cells of the ladder come from re-evaluation, none from transcription. Every one of the 21 endpoints, seven carriers by three splits, improves over its own carrier. On HRSID the two conventional detectors gain +11.9 inshore apiece, from 28.3 to 40.2 for Mask R-CNN and from 30.4 to 42.3 for Cascade R-CNN, and SOLOv2, the weakest carrier in the table at 19.0, gains +14.3 to 33.3; on PSeg-SSDD the three carriers gain +6.9, +16.3, and +6.1. Two intermediate readings are negative and we print them rather than hide them. The training-free rung costs Mask R-CNN 0.6 and YOLACT 0.5 offshore AP on PSeg-SSDD, 65.1 to 64.5 and 63.4 to 62.9, which is the offshore erosion that Section IV-D identified for predictions that are already stable on that split, and the boundary tier converts both to +1.4 and +2.7. The single most useful number in the table is 42.3. Cascade R-CNN, an architecture from 2018 with no SAR-specific component and no orientation prior, lands within 0.8 points of the 43.1 that a specialized 500-epoch design reports on the same inshore split, purely by being made to look eight times and correct itself.[1] These carriers are our own trainings under our own 55-epoch budget, so their absolute levels are not comparable with the 500-epoch rows of Table II; what transfers across the two tables is the delta, not the level.

The size of the gain is not constant, and its variation is itself a test of the diagnosis. If the recipe removes prediction variance, a carrier with more variance left in it should gain more, and carrier strength is a usable proxy for how much is left. The prediction survives on the total gain, and the paragraph after this one shows that its two tiers reach that total from opposite directions. The two datasets separate accordingly. On HRSID, where every carrier alone sits below 31 inshore AP, the recipe returns between +11.9 and +14.8; on PSeg-SSDD, where the three carriers start between 38.9 and 41.2, two of them return +6.9 and +6.1. CondInst is the exception and we treat it as one. It gains +16.3 inshore, the largest figure in the table, from a carrier-alone 41.2 that sits at the top of the PSeg-SSDD group. Across the seven carriers the Spearman rank correlation between carrier-alone inshore AP and inshore gain is -0.77 without CondInst and -0.11 with it, so the tendency holds in six carriers and one carrier flattens it to nothing; we report a tendency and not a law. The exception has a mechanism worth naming. The training-free rung alone is worth +8.9 inshore on CondInst against +3.3 and +2.2 on the other two PSeg-SSDD carriers, so its excess gain sits in the variance tier rather than the boundary tier, which is what per-instance kernels generated from a single view's features should do: they are the most view-sensitive readout of the seven, and the consensus vote consequently finds the most to remove.

The three HRSID carriers sharpen this reading, because they hold dataset, split, schedule, and evaluator fixed and vary only the base, and across them the two error axes scale in opposite directions. The detection-variance axis contracts as the base weakens. The training-free stack returns +8.8 inshore on Mask R-CNN and +8.7 on Cascade R-CNN but +6.5 on SOLOv2, and the same ordering holds on the other two splits, +7.4 and +6.8 against +4.8 official and +5.8 and +4.8 against +2.3 offshore. The consensus vote does not invent detections; it promotes candidates the carrier already produces in some views and suppresses in others, so its yield is bounded by the stock of present-but-suppressed candidates the carrier brings into the fusion, and a base at 19.0 inshore AP brings fewer of them than a base at 30.4. The variance tier therefore meets a floor at the weak end, just as Section IV-J found it running out of material at the strong end, where single-view predictions are already stable; it pays most on the intermediate bases that produce the candidates but rank them inconsistently. The boundary axis runs the other way and runs harder. The refiner adds +3.1 and +3.2 inshore on the two region-based carriers against +7.8 on SOLOv2, +4.0 and +4.3 against +10.0 official, and +4.6 and +5.0 against +11.3 offshore, a factor of 2.3 to 2.5 on every split. SOLOv2 forms each mask by convolving one shared feature map with a kernel predicted at a coarse instance grid rather than through a per-region head at RoI resolution, so its silhouettes are the loosest of the three carriers and its carrier-alone inshore AP_{75} is 15.7 against 24.8 and 27.0, which is exactly the state in which a band-guarded refiner has the most to correct. CondInst, the other kernel-generated readout in the table, behaves the same way on the other dataset with the largest PSeg-SSDD boundary gain at +7.4 inshore, so what drives this axis is the mask readout rather than the base level as such; the two travel together here because one coarse readout produces both the weak carrier and the loose contour. The two trends compose instead of canceling. SOLOv2 still returns the largest HRSID total at +14.3 inshore against +11.9 for both region-based carriers, so the tendency of the preceding paragraph survives at the level of the total, but it is the sum of a shrinking variance term and a growing boundary term rather than one monotone effect. This is the most direct evidence in the article that the two axes are distinct quantities and not two readings of one. Orthogonality by itself leaves their dependence on base strength unconstrained, whereas measured here the two dependences carry opposite signs, which no single-axis account of the near-shore residual can produce.

Fig. 15 shows what these deltas look like on the image. In the 47-ship HRSID harbor of row (a), Mask R-CNN alone

[1] Applying multi-scale flip TTA to Cascade R-CNN first required repairing its mask head. The `CascadeRoIHead` of mmdetection averages the three stages' mask logits through a routine left over from the removed 2.x augmentation-test interface, which consults the `flip` flag and tries to un-flip each stage's RoI-local logits. On a flipped view it raises a negative-stride error, and it would be semantically wrong even if it ran, because within one view all three stages already live in that view's frame and the whole-image flip is undone once, after pasting. Our parameter-free override drops the flip handling and is output-identical to the stock head on non-flipped pipelines.

finds 23 targets and misses 24, and the recipe takes it to 38 found and 9 missed at 20 false alarms against 15. The two intermediate rows behave the same way: Cascade R-CNN on the 95-ship HRSID scene of row (b) moves from 28/44/67 to 54/46/41 true positives, false alarms, and missed targets, and CondInst on the 24-ship PSeg-SSDD scene of row (c) from 6/2/18 to 18/2/6. For each carrier the displayed scene is selected programmatically, maximizing one point per recovered target plus half a point per removed false alarm over scenes carrying at least six annotated ships. In the 88-ship scene of row (d) our own carrier moves from 31 found with 37 false alarms to 71 found with 29, recall and precision improving together, which is the signature the consensus vote produced on our own base in Section IV-H and reproduces here on carriers that share nothing with it except the recipe.

L. Detection-Level Analysis

Mask AP compresses detection and mask quality into a single number, so Table XI re-reads the same PSeg-SSDD predictions as Table IX in the terms an operator uses: how many ships are found, how many false alarms come with them, and how the two trade against each other. This accounting and the curves of Fig. 16 are computed on the refined predictions, before the learned re-ranking of Section IV-G, because the re-ranker rewrites scores and therefore moves what any fixed threshold selects, which would mix a ranking effect into a detection count. Matching is greedy in descending score at mask IoU ≥ 0.5 with one claim per annotated ship. The table reports both a fixed 0.30 score threshold and each method's own F_1-optimal threshold, because that optimum ranges from 0.10 for CondInst to 0.85 for M2F + DenoDet and no single fixed threshold is neutral across that range. Inshore, the refined full method detects 169 of the 172 annotated ships at the fixed threshold ($P_d = 0.983$), against 156 and 159 for M2F + OAEM and M2F + DenoDet and between 113 and 127 for the four generic architectures. The recall is not free: it arrives with 1.41 false alarms per image against 1.11 and 1.00 for the two specialized rows, so the fixed-threshold F_1 of 0.833 sits below the 0.844 of M2F + DenoDet, and we report the fixed-threshold block as a recall-first operating point rather than as a win. At each method's own optimum the ordering resolves on both summary axes, $P_d^* = 0.936$ and $F_1^* = 0.902$ against 0.866 and 0.882 for M2F + OAEM and 0.843 and 0.881 for M2F + DenoDet. Against the generic architectures the comparison is unambiguous at either threshold: the full method finds 168 of 172 inshore ships at 1.44 false alarms per image where Mask R-CNN with deformable convolutions finds 127 at 1.74, more targets and fewer false alarms at the same time.

Offshore, the same accounting is a parity statement and we report it as one. The full method and the two specialized rows all detect 369 of the 374 annotated ships at the fixed threshold, and at their optima all three read P_d^* between 0.979 and 0.984 at 0.038 false alarms per image, so the split is saturated and no method has room to separate. Two entries there need an explicit word. The offshore rows of the full method and of its refined version are identical at every swept threshold, which is the expected behavior and not a transcription error: the refiner reshapes boundaries only inside a ± 4-pixel band, offshore masks already clear the 0.5 matching threshold by a wide margin, and no instance crosses that threshold in either direction. YOLACT's P_d^* of 0.992 is the single best detection rate in the offshore block, but it is reached at a score threshold of 0.15 and at 0.118 false alarms per image, the worst false-alarm figure in that block, which is why P_d and FA/img are printed side by side and why the footnote warns against reading either alone.

Fig. 16 plots the same eight methods as precision-recall curves on the inshore split, at the two IoU thresholds that separate the two mechanisms we claim. At IoU $= 0.50$, which asks only whether the ship was found, the three rows of our pipeline occupy the top of the panel at 90.8 for the base, 94.3 with the training-free stack, and 95.7 with the refiner, against 92.2 for M2F + OAEM, 92.0 for M2F + DenoDet, and 79.1, 72.1, and 71.1 for CondInst, YOLACT, and Mask R-CNN. The generic architectures run out of recall early: the YOLACT and Mask R-CNN curves reach zero precision before a recall of 0.80 and CondInst before 0.90, whereas the refined pipeline still holds a precision of 0.89 at a recall of 0.90 and 0.84 at 0.95, against 0.76 and 0.45 for M2F + OAEM and 0.84 and 0.56 for M2F + DenoDet, and it is the only one of the eight that reaches full recall at nonzero precision. At IoU $= 0.75$ the picture inverts for the training-free tier. The base and the stack read 53.2 and 54.9, below the 55.6 of M2F + OAEM and the 58.8 of M2F + DenoDet, and only the refined row leads, at 72.5. The two panels state the division of labor of the pipeline in one figure: the training-free operators recover detections, the two SAR-specific modules produce slightly tighter masks than our unrefined output, and the boundary refiner is what converts the recovered detections into strict-overlap accuracy.

M. Robustness Across Seeds

The recovery gain is not a single-seed artifact. Applying the six-view msTTA core to each seed individually lifts inshore AP from 27.7/27.6/27.7 to 34.6/35.0/34.4 for s0/s2/s3, respectively: gains of +6.9/+7.4/+6.7, a mean of +7.0 with $\sigma \approx 0.3$, matching the ± 0.3 inter-seed noise band of the base itself. The gain also follows the regime gradient, +7.0 inshore against +5.5 official and +3.4 offshore, largest exactly where single-look speckle and dock clutter make single-view predictions most variable. Every artifact above the seed level is deterministic: the soup is a single fixed checkpoint, the TTA and fusion passes are deterministic inference, the refiner re-run reproduces all reported figures to four decimals, and the re-ranker regenerates its outputs byte-identically (Sections IV-F and IV-G). The recovery framework therefore inherits the seed robustness of its weakest layer, the base training, and adds no stochastic stage of its own.

V. Discussion

A. A Preservation Regularity: When Architecture Engineering Pays

A single regularity organizes the controlled ledger of Section IV: on a strong COCO-pretrained base, a modification

survives in proportion to how much of the pretrained representation it preserves. Every weight-touching intervention we tested failed to pay. Constraining the deformable-attention sampling offsets cost 1.0 inshore point, one-to-many auxiliary supervision was flat, the four training-side families (capacity, feature enrichment, data augmentation, supervision) were each subsumed by the COCO-initialized base, the MaskIoU rescoring head (worth +1.5 inshore standalone) left a +0.1 residual under the weight soup, the zero-gated strip adapter cost 0.9, and the Swin-S backbone swap gained +0.9 at the base yet ended at 37.3/52.5/63.5 under the full recovery stack, below the ResNet-50 configuration on all three splits. The 200-epoch equal-budget control repeated the pattern on the schedule axis, reaching 37.9/52.5/63.5. Meanwhile, the three operators that left the weights untouched delivered the entire +11.1/+8.2/+5.2 lift.

The regularity is regime-dependent, not universal. On a from-scratch Mask2Former at 17.9 inshore mask AP, the specialized modules of O2Former are worth +25.2 points [2]; in our own earlier experiments, a Gaussian–Wasserstein matching cost gained +4.4 inshore (12.5 to 16.9) on an ImageNet-backbone, from-scratch-head base, yet turned neutral to negative (27.9 to 27.4) once the identical model was COCO-initialized. Same mechanism, two regimes, opposite signs. Architecture engineering therefore pays where the representation is weak enough to leave headroom for injected priors, and stops paying once COCO initialization has already supplied what those priors encode. The adaptation literature corroborates this reading from outside SAR: Mona reports that a convolutional adapter preserving the pretrained representation surpasses full fine-tuning by +0.9 mask AP on COCO, whereas the linear LoRand adapter falls 0.4 below it [13]. Preservation is necessary but not sufficient. Our strip adapter respected preservation exactly, through a scalar zero-initialized residual gate that reproduces the COCO model at the start of fine-tuning, and still lost 0.9 inshore point because the elongation prior mismatches the quasi-point-like nearshore hulls of HRSID. The optional boundary-patch refiner is consistent with, not a counterexample to, the same regularity: its 2.35M parameters sit beside the frozen pipeline rather than inside it, the pretrained weight alignment is untouched, and its gain lands precisely on the strict-IoU boundary axis that no weight-touching change bought. The learned re-ranker obeys the regularity even more literally, since it never reads a pixel and never edits a mask, and acts only on the ordering of candidates the frozen pipeline has already produced.

B. What Did Not Help, and Why It Matters

Several directions that should have worked did not, and each failure marks a boundary of the error-axis decomposition rather than a gap in the engineering. The most instructive is speckle marginalization. Multiplicative speckle is the one perturbation that SAR imaging physics specifies exactly, so re-drawing it and adjudicating the resulting decisions is a physically principled fourth view axis, and we archived eight falsifiable predictions before the first GPU run of that arm. Two variants were built and both lost. The first estimates the backscatter field by median despeckling with a bright-scatterer pass-through and re-draws Gamma speckle on that estimate; its best fusion reached 37.1 inshore against 38.8 for the geometric stack and was negative on the other two splits as well, 51.7 official and 62.3 offshore against 53.8 and 64.8, while its base-free majority vote fell to 27.0, below the 29.6 of plain single-view inference, which convicts the despeckling estimator of erasing 10- to 30-pixel hulls. The second variant drops that estimator and multiplies mild Gamma draws onto the image directly; its base-free vote recovers to 30.0, clearing the estimator of blame, yet the fused result still stops at 37.3. The reason is subset absorption. Multi-scale inference already resamples every chip between 640 and 1088 pixels, which alters the effective speckle correlation, so the decisions a speckle re-draw can flip lie inside the set the geometric views flip. The gain theorem survives untouched; what does not survive is its empirical premise that an unharvested speckle reserve exists on a base trained with heavy scale jitter. One prediction did survive, and was confirmed twice: fusion that keeps the base prediction as an arbiter beat pure K-draw majority voting in both variants, 37.1 against 27.0 and 37.3 against 30.0, exactly as the negative-margin analysis warned that marginalization without an arbiter would freeze confident errors in place. A theory that predicts the shape of its own failure is worth more than one that predicts only successes.

The architecture side closed the same way. A minimal-footprint variant of the base improved every split of both datasets, reaching 28.2/46.3/60.5 on HRSID against 27.7/45.6/59.6 for the audited base, and then surrendered the entire improvement the moment the stack was applied to it: 36.4/51.6/63.0 against 38.8/53.8/64.8 for the unmodified base under the identical eight-view protocol, carrying the single-seed, soup-free caveat of the Swin-S arm. A stem lateral connection cost 6.9 AP_{75} points outright, and a gentler resampling variant, positive in aggregate, still gave up 1.0 inshore AP_{75}, the very axis it was built to serve. Together with the Swin-S and 200-epoch controls, this is the direct form of the interchangeability claim: a training-side gain and an inference-side gain that act on the same detection-variance axis substitute for one another instead of adding, because the stack has already driven that axis toward its floor and a better-trained base merely reaches the same floor from below. Two further nulls are of a different kind. A miss-rescue verifier that re-scores the below-threshold reject pool moved the three splits by at most 0.03 AP, because the training-free stack has already lifted every recoverable ship out of that pool. A 16-feature re-ranker, built by adding decoder-depth consistency to the 14 consensus and geometry statistics, tied with the smaller model on all three splits, 44.5/59.3/70.0 against 44.5/59.4/70.1, because stability across decoder depth and agreement across views report the same evidence twice. Read as a ledger of failures this is discouraging. Read as a map it is not: every axis we could name has exactly one remedy that moves it, and the directions that fail are the ones duplicating a remedy already in place.

C. Limitations

Four limitations bound our claims. First, single-look amplitude imagery imposes a sensor ceiling. The physics that separates a berthed hull from the dock structure it touches, including azimuth sub-aperture anisotropy and polarimetric signatures, is absent from the input, and a radiometric-consistency re-ranking probe recovered +0.0 inshore over a flip-TTA control, consistent with, though not proof of, a modality-bounded residual. Second, the residual gap has moved rather than closed. The near-shore split of HRSID is now ours by 1.4 AP and 2.4 AP_{75}, 44.5 and 48.3 against 43.1 and 45.9 for O2Former [2], so the strict-overlap axis on which specialized architectures held their near-shore advantage is no longer a deficit for us; offshore, however, still trails by 1.6 points (70.1 versus 71.7, and 70.9 for DiffSARShipInst [3]), the PSeg-SSDD inshore split remains 1.1 below the same architecture (62.1 versus 63.2) and its offshore split 2.1 below (69.8 versus 71.9), and the training-free tier alone does not lead on absolute inshore AP (38.8 versus 43.1). Third, the evidence base is single-GPU and two-dataset: the recipe transfers from HRSID to PSeg-SSDD with every operator's sign preserved (Section IV-J) and leaves five frozen carriers we did not design above their own baselines on every split, but both datasets are single-sensor amplitude collections, and the architecture- and training-side sweep is eliminative rather than exhaustive, so stronger interventions than the forms we tested may exist. Fourth, two of the three tiers are trained, so every training-free claim is scoped to the recovery stack of Section III. The boundary-patch refiner (2.35M parameters, 19 minutes) and the consensus re-ranker (14 scalar features, 19 seconds) are reported as separate tiers in every table, and each is refitted on the training split of the dataset it is applied to, which makes their cross-dataset transfer a re-fit rather than a zero-shot carry-over; the refined-tier comparison also inherits the protocol caveat of Section IV, where 55 COCO-initialized epochs and 500 from-scratch epochs are different operating points. Within these bounds, the central results stand on same-ground-truth controlled comparisons: a recovery of +16.8/+13.8/+10.5 points over the audited base, an inshore AP_{50} of 75.6 from the training-free tier alone against 69.5 for the strongest published architecture, and near-shore AP and AP_{75} above that architecture at roughly a tenth of its training epochs.

VI. Conclusion

This article diagnosed the near-shore residual of a strong COCO-pretrained Mask2Former on HRSID as a variance problem rather than a capacity problem, with the pretraining regime itself the largest single lever: 17.9 versus 27.7 inshore mask AP for the identical architecture. We introduced a training-free recovery stack, composed of a same-basin weight soup, wide multi-scale flip test-time augmentation, and consensus-driven mask-IoU vote fusion, which lifted the base from 27.7/45.6/59.6 to 38.8/53.8/64.8 inshore/official/offshore mask AP at zero additional training and roughly $8\times$ inference cost. On top of that stack we devised two small correctors, each aimed at one unsaturated error axis and each supervised by nothing but the frozen model's own mistakes on the training split: a 2.35M-parameter boundary-patch refiner, band-guarded to a ± 4-pixel strip around predicted contours and audited along seven leakage vectors, and a gradient-boosted re-ranker over 14 consensus and geometry statistics that reorders detections after 19 seconds of fitting without altering a single mask pixel. The three-tier cascade reached 44.5/59.4/70.1 mask AP with an inshore AP_{75} of 48.3, placing near-shore accuracy 1.4 AP and 2.4 AP_{75} above the strongest specialized 500-epoch architecture at roughly a tenth of its training epochs. Controlled negatives carried the diagnosis. Architecture- and training-side directions on the strong base were flat, negative, or absorbed by the stack that followed them, a 200-epoch equal-budget control did not close the gap, speckle marginalization failed in the manner its own theory had predicted, and the fusion ablation showed that the merge operator, not the view count, decides whether aggressive test-time augmentation helps or collapses. Universality was measured rather than asserted. On five frozen carriers we did not design, spanning two-stage region-based, cascade, dynamic-kernel, prototype-based, and grid-based mask-kernel designs, across both datasets and all three splits, every carrier ended above its own baseline, the training-free and boundary tiers proved to depend on carrier strength with opposite signs, and Cascade R-CNN carrying the recipe reached 42.3 inshore mask AP, within 0.8 point of an architecture purpose-built for this task. Transferred to PSeg-SSDD with no change beyond a proportional rescaling of the augmentation scales, the complete recipe took our own base from 51.9/60.9/64.8 to 62.1/67.5/69.8. In the future, we plan to extend the recipe to sensors providing polarimetric or sub-aperture information, where the amplitude ceiling identified here may be broken. What an operator receives from this work is not a stronger model but a setting: on strong pretrained bases the frontier has left the weights, and what remains is to decide how much deliberation each image deserves.

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